The Limitations of Equilibrium Concepts In Evolutionary Game Theory

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Heartfelt thanks to
Kevin, for your guidance, support and laughter,
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Abstract

Within the modeling framework of evolutionary game theory static equilibrium concepts adapted from rational choice game theory are employed to identify the probable outcomes of dynamic evolutionary processes. Over the last several decades, results have emerged in the literature demonstrating limitations of each of the proposed equilibrium concepts.

We present a comprehensive story circumscribing the shortcomings of the Nash equilibrium, evolutionarily stable strategy, neutrally stable strategy, and evolutionarily stable set concepts. We argue that these results rely on an implicit notion of evolutionary significance, and propose a novel account of evolutionary significance. We show how this formulation brings clarity to assessments of the success and failure of equilibrium concepts across different models of evolution.

We demonstrate that even under the most favorable assumptions regarding the underlying dynamics and conditions for stability—those of replicator dynamics, and asymptotic stability—each equilibrium concept falls short. Each one is simultaneously too weak, and too strong. That is, each picks out population states that we would not consider plausible outcomes of an evolutionary process, and each fails to pick out states that constitute eminently plausible evolutionary outcomes.

We extend our analysis to the standard stochastic model of evolution—the Moran process—where we derive a suitable formalization of evolutionary significance for the stochastic context and use this to assess the performance of the equilibrium concepts for different parameterizations of population size, and frequency of mutation.

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1 Introduction

The search for an equilibrium concept in the theory of evolutionary games (EGT) has been driven by the hopes for an effective, general, and theoretically simple method of analysis. Forty years since Maynard Smith and Price's introduction of the concept of the evolutionarily stable strategy (ESS) [1973], it seems that no such equilibrium concept is to be had, and this absence has become a part of the folk wisdom of the community of practitioners of EGT. Yet, in certain areas of theoretical biology, it is not uncommon to find the exclusive use of the equilibrium concepts, the leading candidate among which is the still the ESS.

A search for "Ecology" and "ESS" in Google Scholar reveals that 13 of the 20 top papers since 2013 that used the method of ESS analysis used it exclusively. That is, they followed what Huttegger and Zollman have termed "ESS methodology" [2014]. These papers describe an evolutionary phenomenon in terms of a game, find the ESS or ESSs of that game, and equate these with the possible outcomes of evolution. This can be problematic. As we will see, there are a breadth of conditions under which each of the primary equilibrium concepts make serious errors of both omission and commission. Indeed, there are in-principle reasons why we should expect equilibrium concepts to come up short. This thesis presents a coherent story of how and why this is so.

In §1.1, we consider the basic anatomy of evolutionary games, and introduce the concepts of games and evolutionary dynamics needed to understand our methodology and results. In §1.2, we unpack the notions that we argue implicitly underwrite the informal goal of capturing "evolutionary significance" until we reach a formulation that lends itself to being made mathematically precise, and that we can then use to guide our evaluations of the success and failure of equilibrium concepts. In §1.3, we present the stability concepts that, with respect to a given family of dynamics, attempt to formalize our intuitions about evolutionary significance. In §1.4, we introduce a precise characterization of equilibrium concepts, and examine their relationship to our stability concepts for evaluative purposes.

In §1.5, we explain the methodology and assumptions underlying the results we will examine. We defend the use of replicator dynamics (RD) and asymptotic stability (AS) as our models of evolution, and conditions for evolutionarily significant outcomes, respectively. We argue that RD and AS constitute assumptions strongly favorable to the success of the equilibrium concepts, and so their failure with respect to these assumptions provides strong results against them.

In §2.2, we move on to circumscribe the limitations of the main equilibrium concepts used to analyze evolutionary games. That is, we consider the Nash equilibrium (NE), strict Nash equilibrium (strict NE), evolutionarily stable strategy (ESS), neutrally stable strategy (NSS), and evolutionarily stable set (ES set) concepts. We show how each equilibrium concept is simultaneously too weak, and too strong. That is, we show that each one picks out population states that we would not consider evolutionarily significant, and each one fails to pick out what are clearly evolutionarily significant outcomes. We also provide a distinct in-principle argument for expecting that any point- or set-valued equilibrium concept will fail in capturing all significant outcomes.

In §2.3, we consider how the limitative results fair under the alternative assumption of Lyapunov stability. We discuss the results with an eye to their implications for the fate of equilibrium concepts, and the challenges faced by any viable method of analysis in capturing our intuitive notions of evolutionary significance.

In §3, we extend our analysis of the limitations of equilibrium concepts to stochastic models of evolution. We introduce the Moran process (MP) and the Moran process with mutation (MPM), and present new methods needed for their analysis. In §3.2, we use our novel account of evolutionary significance to select viable stability concepts for the stochastic context, and to enable meaningful comparison of our results between the deterministic and stochastic contexts.

In §3.3, we circumscribe the limitations of equilibrium concepts under the Moran process in the limit of large population size, and the Moran process with mutation in the limits of large population size and vanishing mutation rates. We demonstrate how the equilibrium concepts fair both worse and better under these assumptions. We then examine the fate of equilibrium concepts under the more realistic assumptions of finite population size and nonzero mutation rates. An interesting and unexpected result emerges indicating that strong mutation may provide a sufficient condition for substantive agreement between the Moran process and the replicator dynamics. In §3.4, we discuss the implications of our results. In §4, we conclude.

1.1 Evolutionary Games and Dynamics

We begin by defining an *evolutionary game* E as composed of two parts: a *game* G, and an underlying *evolutionary dynamics* D. The game describes the possible interactions between individuals using various strategies. Evolutionary games are used in many contexts—e.g. animal behavior, ecology, prebiotic evolution, epidemiology, oncology, economics, and anthropology—where players can be interpreted variously as representing genes, individuals (bacteria, people, and so on), or memes.

For simplicity, we stick to a largely biological interpretations and vocabulary where players are to represent individual organisms. Strategies are to be understood as behavior patterns or phenotypes, and payoffs as Darwinian fitness or reproductive success (i.e. expected number of offspring). We can describe a game using a matrix A specifying the possible outcomes of interactions between combinations of strategies in terms of the resulting fitness of each strategy involved. For example, consider the generic two-player symmetric game given in $Figure\ 1$. If one member of the population uses strategy S_1 against an individual using strategy S_2, then the matrix specifies that their resulting finesses are $A(S_1, S_2) = a_{12}$ and $A(S_2, S_1) = a_{21}$ respectively.

$$A = \begin{array}{c|cc} & S_1 & S_2 \\ \hline S_1 & a_{11} & a_{12} \\ S_2 & a_{21} & a_{22} \end{array}$$

Figure 1

There is an interesting ambiguity as to how to interpret mixed strategies (see Ch.3 of Osborne and Rubinstein [1994]). Formally, a mixed strategy is a weighted combination of strategies with the weights summing to one. Looking at the two strategy game presented in $Figure\ 1$, a mixed strategy is any strategy $S = x \cdot S_1 + (1 - x) \cdot S_2$ where $x \in [0,1]$. In the context of an evolutionary game, S may be interpreted as reflecting a monomorphic (single type) population where individuals perform different actions with probabilities x and $(1 - x)$, or as reflecting a polymorphic (multi-type) population where the mixture reflects the proportion of the population playing each of the available pure strategies S_1 and S_2. For simplicity we stick with the latter interpretation throughout this thesis.

More generally, a symmetric normal form two-player game G is specified by a quadruple $\langle P, S, \Delta^n, A \rangle$ composed of: the set of *players* $P = \{p_1, p_2\}$ involved in each interaction; the set of *pure strategies* $S = \{e_1, \dots, e_n\}$ possible for the players to perform; the *state space* $\Delta^n = \{\langle x_1, \dots, x_n \rangle \mid x_i \geq 0,\ \sum x_i = 1\}$ of all possible combinations of strategies, where each $x \in \Delta^n$ denotes a *population state* (a point in the state space), and a *payoff matrix* $A = [a_{ij}]$ specifying the resulting payoffs to

each player from each combination of strategies. The *expected payoff* to a player using strategy x in a contest with a player using strategy y is then given by $u(x, y) = \sum_{i,j=1}^{n} x_i A(e_i, e_j) y_j = \sum_{i,j=1}^{n} x_i a_{ij} y_j$.

For the results considered here, we restrict our interest to a relevant class of games. Namely, we consider *finite non-cooperative games,* where interactions are between a finite numbers of individuals, there are a finite number of possible (pure) strategies, and individuals are assumed to have at least partially conflicting interests. We consider both simultaneous and sequential games. For the examples in this chapter, we will typically use symmetric normal form two- and three-strategy games.

The second component of an evolutionary game is the evolutionary dynamics. The evolutionary dynamics translate the resulting payoffs of these interactions between strategies into changes in the proportions (or numbers) of the population using the different strategies. The leading idea is that, with the introduction of a dynamics, the proportions (or numbers) of the strategies in the population affect the payoffs to each strategy, and so, fitness becomes *frequency dependent*.

The dynamics describe how evolution behaves, but they may bring along very different sets of modeling assumptions. The dynamics may be discrete, and broken up into repeated rounds of playing the game and updating the population proportions (or numbers). Or the dynamics may be continuous, describing an idealized, fluid process of evolution. The dynamics may be deterministic, so that any identical population with identical initial conditions will evolve identically for all time. Deterministic dynamics are often described by systems of differential equations. Or the dynamics may be stochastic, so that the populations evolve probabilistically, and populations starting from identical initial conditions can go on to trace distinct paths through the state space. Markov processes are often used describe stochastic dynamics.

The dynamics also typically define the population size and structure. This includes whether the population is taken to be an infinite unit mass, or composed of a finite number of individuals, and whether one type of individual is more or less likely to interact with another kind, or whether interactions are wholly random. Reproduction may be sexual or asexual. Transmission of behavior, or genes may be perfect, or there may be the possibility of mutations occurring. Drift occurring in the absence of selective forces may be accounted for, or not. Finally, other features of the model that describe neither evolution nor the population—such as environmental variability—will also typically be introduced as a part of the dynamics.

For the results considered in this chapter, we employ Taylor and Jonker's replicator dynamics (RD) [1978]. We will discuss the methodological justification for the choice of RD in §1.5. For now, we observe that in their simplest form, RD assume: determinism, an infinite populations, random pairwise interaction, non-overlapping generations, asexual reproduction, perfect transmission of strategies from one generation to the next, a corresponding absence of mutation, the absence of drift, and a constant environment.

RD can be formulated to treat time as discrete or continuous. The discrete case can make it clear how RD works, so we consider it first. The leading idea of RD is that the growth rate of each strategy is a function of the difference of the strategy's fitness and the population's mean fitness. Formally, discrete time RD consists of a mapping $f: \Delta^n \times A \to \Delta^n$ that is specified by a system of difference equations

$$f_i(t+1) = f_i(t) \cdot \frac{U(i,f)}{U(f,f)} \text{ for each strategy } i = 1, \ldots, n.$$

Here $f_i(t)$ denotes the proportion of strategy i at time t, $f_i(t+1)$ denotes the proportion in the next time step, and f is shorthand for $f(t)$ and denotes the makeup of composition of the population as a whole.

The payoff to an i-type against the population is the sum of the payoffs it receives against each strategy weighted by their proportion in the population, yielding $u(i,f) = \sum_j^n f_j(t) \cdot A(i,j)$ for $j = 1, \ldots, n$. Similarly, the fitness of the population against itself is given by $u(f,f) = \sum_j^n \sum_k^n f_j(t) \cdot A(j,k)$. Thus, we can more explicitly formulate the difference equations as

$$f_i(t+1) = f_i(t) \cdot \frac{\sum_j^n f_j(t) \cdot A(i,j)}{\sum_k^n \sum_j^n f_j(t) \cdot A(k,j)}.$$

Taking this discrete process, we can reformulate it to represent the change in the population at any given time, and so derive a continuous dynamics [Taylor and Jonker, 1978]. Continuous time RD consists of a flow that may be specified by a system of differential equations

$$\frac{df_i}{dt} = f_i(t) \cdot \frac{u(i,f) - u(f,f)}{u(f,f)}.$$

where $\frac{df_i}{dt}$ is the rate of change of the population proportion of strategy i at time t. The normalizing term $u(f,f)$ is typically dropped for ease of computation (as we

will do from this point forward). We note that an equilibrium point is just a *stationary point* of the dynamics. Formally,

> **Definition 1.1.1 (Stationary Point in Discrete RD)** In discrete time replicator dynamics, a state $f(t)$ is a *stationary point* if $f(t+1) = f(t)$.

> **Definition 1.1.2 (Stationary Point in Continuous RD)** In continuous time replicator dynamics, a state $f =< f_1, \dots, f_n >$ is a *stationary point* if $\frac{df_i}{dt} = 0$ for all i.

For our analysis, important properties of RD to bear in mind are that it is monotone, autonomous, and forward invariant. Intuitively, monotonicity means that strategies with higher fitness have higher growth rates. Formally, a dynamic f is *monotone* if for all $x, f_i(x)/x_i > f_j(x)/x_j$ just in case $u(i,p) > u(j,p)$. For a dynamic to be *autonomous* means that, for determining future states of the system, the past can be considered irrelevant given the present. *Forward invariance* is the property of having uniquely determinable trajectories. That is, for every initial state x at $t = 0$, there is a unique trajectory $x(t) \in \Delta$ for all $t \geq 0$. For further examination of discrete and continuous time formulations of RD see [Schuster & Sigmund, 1983]. The results we consider here employ continuous time RD exclusively.

1.2 A Novel Account of Evolutionary Significance

To appreciate the results discussed in this thesis, we need an explicit understanding of how to evaluate the success of equilibrium solution concepts in the context of evolutionary games. This is often explained in terms of the idea of *evolutionary significance*. That is, it is assumed that for an equilibrium concept to be successful it must capture all and only the states of an evolving population that are evolutionarily significant [Huttegger and Zollman, 2010].

Our challenge is that evolutionary significance is an essentially informal concept. The notion of evolutionary significance has been used a number of times in the EGT literature [Skyrms, 2000; Zollman et al, 2012; Huttegger et al, 2014], but has never been explicitly defined. We attempt to provide such a definition in the context of evolutionary games, and in doing so, arrive at a novel probabilistic account of evolutionary significance. Our definition remains true to the formal framework of EGT, while sketching clearer connections between the intuitive

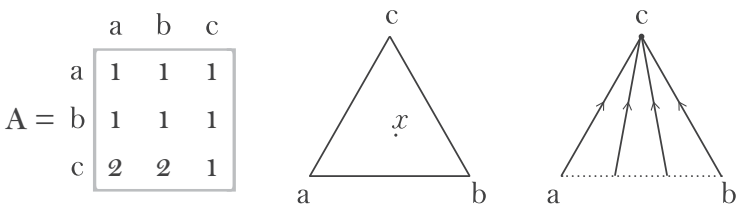
notion of evolutionary significance, the formal definitions of stability concepts, and equilibrium concepts.

To begin, what we want to say is something to the effect that the significant outcomes of an evolutionary process are just those that are *the probable outcomes of evolution*.

> **Definition 1.2.1 (Evolutionarily Significant Outcome)** In an evolutionary game, an *outcome* is *evolutionarily significant* if, and only if, it is a probable outcome of evolution.

But this definition needs be unpacked and sharpened in turn, and we will continue a process of unpacking and sharpening the assumptions implicit in evolutionary significance until we arrive at those amenable to precise formulation. The result of this process will be to provide us with the criterion that we will use in evaluating the success of equilibrium concepts. We begin by considering our notions of 'evolution', and 'probable outcome'.

To illustrate the concepts involved here, we use the following evolutionary game that exhibits a single probable long run outcome. Let the simplexes in $Figure\ 2$ be the state space of a simple 3-strategy symmetric evolutionary game. That is, each point in the simplex represents a possible population state—a convex combination (a mixture with proportions summing to one) of three strategies in the game. This will be represented by a state vector $x = \langle x_1, x_2, x_3 \rangle$ specifying the population proportions of the strategies a, b, c respectively. The three vertices represent the monomorphic states where the population is composed entirely of a single strategy, so the a-vertex corresponds to the state vector $\langle 1, 0, 0 \rangle$, and so on. Distance from a vertex corresponds inversely to the proportion of the strategy represented by the vertex in the population.



Matrix A Simplex I Simplex II

Figure 2: Single Probable Outcome

$Matrix\ A$ above details how each strategy in the population fairs when faced with any other strategy, including itself. We see that an individual using strategy c and facing an individual using strategy a will receive a payoff $A(c, a)$ of 2. Given a game, the dynamics specify how the state vector changes over time, moving it

through the state space of possible population proportions. The path that a population moves along in the state space as it evolves is called its trajectory. Which trajectory a population takes depends on its initial conditions. We visualize the "flow" created in the state space by the dynamics (which we can do in this case, as our game has only three pure strategies) by drawing representative trajectories where the directions of flow are designated by using arrows as in $Figure\ 2$. Diagrams, such $Simplex\ I, II$ in $Figure\ 2$ that describe our game dynamics are called phase portraits. Here, dotted lines represent sets of stationary points, opaque circles represent sinks, and empty circles represent sources. These concepts will be explained in detail at the beginning of §3.

In this game, we would like to say that the state in which strategy c dominates the population is the sole probable outcome of evolution. The reasoning is as follows. We begin by assuming that it is equiprobable that a population will start at any particular state, that is, at any point in the simplex. Given this, we can assign a uniform (or continuous or exponential) probability distribution over every point in the space with respect to it being the initial condition of the dynamic.

Here, we observe that the idea of evolutionary significance takes on meaning in terms of a given population and model of evolution. We also note that we have already implicitly assumed a model of evolution—the replicator dynamics. Recall that it is implicit in RD that the role of mutation is negligible. Thus, there is no evolutionary "innovation" in the sense that anytime a strategy is not represented in the population (has a proportion of 0) it will remain so from then on. By introducing the evolutionary dynamics that describe our assumptions as to the basic features of our population and the behavior of evolution, we get the phase portrait in $Simplex\ II$.

We observe that from *most any* starting point, the dynamics will move toward the top vertex of our simplex, representing the monomorphic state where the population is composed entirely of individuals using strategy c. More precisely, we observe that from any state where the proportion of strategy c is nonzero, the dynamics, given enough time, will converge to the state where the population proportion of strategy c is one, and will remain at that point thereafter. So the state $x^* = \langle 0,0,1 \rangle$ where only strategy c is present is a probable outcome of evolution in this game.

We further develop our concept of "probable outcome" by considering an example of an *improbable outcome*. Consider the dotted edge $\overline{ab}$ at the bottom of the simplex. This line corresponds to the mixtures of a and b where strategy c is wholly absent. At all population states along this line, excepting its endpoints,

strategies a and b receive the same fitness. There is no selective pressure in either direction because

$$U_a = x \cdot A(a,a) + (1-x) \cdot A(a,b) = 1, \text{ and}$$
$$U_b = x \cdot A(b,a) + (1-x) \cdot A(b,b) = 1$$

where x and $(1-x)$ are the proportions of strategies a and b, respectively. Given that the fitness of both strategies is equal, it seems that a population starting at this edge would have no reason to leave, and so would remain here for all time.

From a probabilistic standpoint, however, the chances of actually starting on this edge are negligible, since any one-dimensional edge composes a measure-zero subset of a two-dimensional space. By this reasoning, the edge $\overline{ab}$ constitutes an improbable initial condition of the population, and so, not a probable outcome of our dynamic.

But what if our priors are not uniformly (or continuously) distributed along the whole state space? Imagine that we know that the population will start at one of the population states represented by the edge $\overline{ab}$. In this case, we may ask again: does any portion of the edge compose a probable outcome of our evolutionary game? Perhaps surprisingly, our conclusion remains that no subset of the edge is a probable outcome of evolution. The reason for this is a "hidden mutation assumption." That is, while it not explicitly model in the replicator dynamics, we assume that there is some possibility of mutations occurring. This is often motivated by the thought that small numbers of any alternative strategy may be produced by mutation, and lead to small but persistent fluctuations in the population proportions. Here, any small perturbations of the population state from the edge would cause the population to head off toward the top of the state space, never to return. The inevitable presence of mutant strategies and natural fluctuations means that an outcome cannot be a probable one unless it is stable under arbitrarily small perturbations.

We now have a definition of probable outcome in terms of convergence from a significant set of initial conditions, and resistance to small perturbations.

> **Definition 1.2.2 (Probable Outcome)** An outcome is *probable* if, and only if, given a uniform (or continuous or exponential) probability distribution over initial conditions of the state space, a significant (non-measure-zero) set of initial conditions converge to the outcome, and upon convergence are stable in the face of arbitrarily small perturbations.

But we must also be precise about what constitutes an *outcome*. In order to provide a principled and unifying formulation of what constitutes an outcome of a nonterminal process, we must lean on the mathematical notion of limits. An outcome becomes a possible behavior of the system in the limit as time goes to infinity. Thus we have

> **Definition 1.2.3 (Outcome)** An *outcome* is a possible behavior of an evolutionary game in the long run (as time goes to infinity).

But there something more we want to say. We may want to include the normative qualification that an outcome also be an *optimally informative description*. In our example, the behavior of the system stays at a single point, but it need not be so. In the rock paper scissors (RPS) game, the limit behavior of the dynamics is a continuum of cycles around the center of the state space (See *Figure* 3), and any point on the interior of the simplex is a part of some cycle for some initial conditions of the system. In this case it is accurate, but less than ideally informative, to submit that the entire simplex constitutes the probable outcome of the game. A more informative answer can be stated clearly and concisely in qualitative terms: The dynamics will settle into one of a continuum of cyclical orbits around the center point x^* of the simplex.

With the introduction of complex limit behavior like cycles, chaotic trajectories [Skyrms, 1992], and other disequilibrium behavior, some outcomes may be best described in qualitative terms. However, extant equilibrium concepts respond to the question of "what are the evolutionarily significant outcomes?" exclusively in terms of states or sets of states. The observation that there exist important evolutionary outcomes not effectively expressible as sets of points indicates an in-principle argument against equilibrium solution concepts that we will return to in §2.2.

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

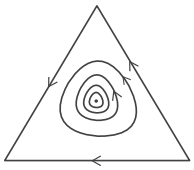


Figure 3: RPS Game

Unfortunately, there is, as of now, no formal classification or qualitative typology of outcome types that are not amenable to being defined in terms of (sets of) states. But we will make do here by limiting ourselves to evolutionary outcomes uniquely definable in terms of (sets of) states. We can still arrive at decent formal conditions by which to evaluate equilibrium concepts. What this means however, is that while the failure of an equilibrium concept to capture an evolutionarily significant outcome is a clear mark against it, the prediction of states that are not evolutionarily significant (by our working definition) does not necessitate an error

on the part of the equilibrium concept. It is logically possible that the equilibrium concept may capture evolutionarily significant outcomes wrongly excluded by our stipulation that outcomes be definable in terms of sets. To assess such cases, we must refer back to our less precise intuitive notions of evolutionary significance to determine whether the equilibrium concept is indeed in error.

Having limited ourselves to outcomes uniquely definable in terms of (sets of) population states, we are interested in the subset of outcomes that are probable (as previously defined). At this point we are in a position to combine definitions 1.2.2, and 1.2.3 to get:

> **Definition 1.2.4 (Probable Outcomes in terms of [Sets of] States)** A set of states is a *probable outcome* if and only if, given a continuous (or exponential) probability distribution on initial conditions of the state space, a significant set of initial conditions converge to the set in the limit, where they are stable in the face of arbitrarily small perturbations.

We have characterized evolutionary significance in terms of probable outcomes under some evolutionary dynamics, and probable outcomes as states that induce convergence from a significant set of initial conditions, and exhibit stability under perturbations. This is what we want, as we can mathematically formalize the concepts of *convergence* and *stability* in the form of our stability concept, and we can then use the stability concept to evaluate the success and failure of equilibrium concepts. For our results in this section, this will end up being asymptotic stability. We note that the condition that a system stays at a state after convergence may be weakened to staying within a fixed distance of the state. This weakened stability condition is called Lyapunov stability, and will also be considered as an alternative assumption in §2.3.

Having arrived at a sharper definition of evolutionary significance for this context, we can return to the simple evolutionary game in *Figure I* to conclude that the single probable outcome of this evolutionary game is the monomorphic population state $x^* =< 0,0,1 >$ where strategy c has come to fixation. The state x^* is evolutionarily significant because it commands convergence from a probabilistically significant set of initial conditions composed of the interior of the simplex, and any population already at x^* is stable. There are no other evolutionarily significant (sets of) states, as any points not on in the interior of the simplex is on its edges, and the edges of the simplex compose a measure-zero subset of the state space, and so do no constitute probable initial conditions.

Under conditions as simple as the ones just described, the equilibrium solution concepts we will consider do indeed pick out x^* as evolutionarily significant, and the ESS & ESSets concepts will pick it out as the sole outcome of this game. As we will see however, when the interactions in our games become even slightly more complicated, there arise conditions under which the equilibrium concepts unequivocally fail to capture our notions of evolutionary significance.

1.3 Stability Concepts

We have provided a sharper definition of evolutionary significance so that we can have clarity about the connection between our intuitive notion of evolutionary significance and mathematical precise formulations of this notion. These formulations are *stability concepts*.

First, we note that a basic fact about stability concepts is that they are dynamics-specific. Stability concepts are defined in relation to dynamics, or classes of dynamics. A stability concept for stochastic dynamics is unlikely to work for deterministic dynamics, as the former is phrased in a language of probabilities, and the latter in a language of certainties.

For our limitative results, we will define the stability concepts for our designated dynamic—the replicator dynamics. The main candidate stability concepts then are *asymptotic stability*, and *Lyapunov stability* [Weibull, 1995]. Intuitively, a state is Lyapunov stable (often shortened to stable) if states near it remain within a fixed distance of it, and a state is asymptotically stable if states near it converge to it. Asymptotic stability is a strengthening of Lyapunov stability with the requirement of *attraction*. We focus on asymptotic stability, and return to consider the results of relaxing our standards for evolutionary stability to Lyapunov stability in the discussion.

> **Definition 1.3.1 (Lyapunov Stable)** A state is *stable* if points near it remain near it. Formally, A stationary point $x^* \in \Delta$ is stable for an autonomous dynamic if for all open neighborhoods U of x^*, there is another open neighborhood O of x^* such that $O \cap \Delta$ remains inside $U \cap \Delta$. A state $x^* \in \Delta$ is *unstable* if it is not stable.

> **Definition 1.3.2 (Attracting)** A state $x^* \in \Delta$ is *attracting* if nearby points tend toward it. Formally, a stationary point $x^* \in \Delta$ is attracting for an autonomous dynamic if there is some open

neighborhood U of x^* such that all trajectories that are in initially in $U \cap \Delta$ converge to x^* in the limit. In this case, $U \cap \Delta$ is called a *basin of attraction* of x^*.

Definition 1.3.3 (Asymptotically Stable State) A stationary point is *asymptotically stable* if it is both stable, and attracting. [Cressman, 2003]

The stability concept provides criteria for evolutionary significance in relation to the game dynamics. By being defined in terms of the dynamics, the stability concept provides a richer, more reliable but less general notion than that can be provided by an equilibrium concept. It becomes clear then, that if the underlying dynamics is taken seriously, the goal of the equilibrium concept should be to pick out essentially the same (sets of) states as the stability concept.

1.4 Equilibrium Concepts

Equilibrium concepts were imported into evolutionary game theory from classical, rational choice game theory. An essential insight that has motivated the adaptation of equilibrium concepts into evolutionary game theory is that evolutionary processes can be thought of as solving optimization problems in a way not entirely dissimilar to the recommendations made to rational agents in classical game theory. Where strategies maximizing individual expected utilities may deliver a group of rational agents to an equilibrium, evolutionary dynamics driven by differentials of fitness expressing reproductive (or imitative) success may deliver populations of nonrational, or bounded rationality agents, to stable states.

The candidate equilibrium concepts are: Nash equilibrium (NE), strict and pure Nash equilibrium, evolutionary stable strategies (ESS), neutrally stable strategies (NSS), and evolutionarily stable sets (ESSets). In the context of evolutionary games, each of these equilibrium concepts aims to capture all and only the evolutionary significant outcomes. Moreover, each is a refinement of the Nash equilibrium. That is, each starts with the set of Nash equilibria of a game and trims down the set by eliminating those equilibria it deems implausible [Huttegger & Zollman, 2014]. The exception to this is the ESSet concept. Rather than assessing individual equilibrium states, the ESSet picks out sets of states that, when considered collectively, exhibit equilibrium properties.

Formally, an equilibrium concept is a mapping F from the class of all games Γ to the union of the states spaces of all possible strategy profiles.

$$F: \Gamma \to \bigcup_{G \in \Gamma} 2^{\Delta_G}$$

That is, F takes as arguments a game G, and the set of the possible states Δ_G for that game (its state space as previously defined), and outputs a subset $F(G, \Delta_G)$ of those states. The goal being, that this subset turns out to be just the evolutionarily significant states.

What is crucial to observe is that: whereas stability concepts are defined in relation to the evolutionary dynamics, equilibrium concepts are defined exclusively in relation to the interaction structure of the strategies. That is, equilibrium concepts are defined exclusively in terms of the game. That equilibrium concepts attempt to infer the plausible outcomes of evolution by considering only the game matrix—and without deferring to any specific underlying model of the evolutionary dynamics—is essential to both their theoretical simplicity and generality, and, ultimately, to their limitations and unreliability.

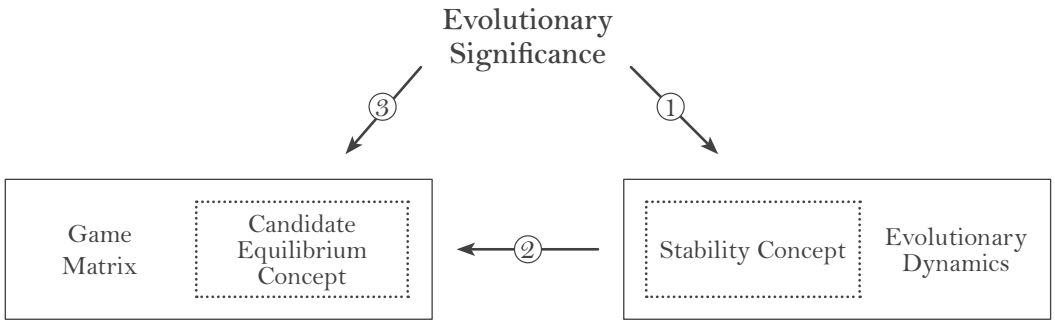


Figure 4: Assessment Diagram

In *Figure* 4, the interrelationship of evolutionary significance, equilibrium concepts, and stability concepts is visualized. In (1), our informal notion of evolutionary significance must be made mathematically precise in the form stability concept, which must be defined with respect to some (class of) evolutionary dynamics. In (2), we can now evaluate the equilibrium concept, defined at the level of the game matrix, in terms of its agreement with the stability concept. This gives us, in (3), a way to indirectly assess the success of a given equilibrium concept in capturing evolutionary significance under a given class of matrices and dynamics. For us, these are finite games under the replicator dynamics.

In this way, equilibrium concepts explain evolutionary significance in so far as they are in agreement with the stability concept; we assess each equilibrium concept by its success in capturing the set of states that are asymptotically stable.

We have presented a novel, and more precise, formulation of evolutionary significance that we argue captures what implicitly underpins determination of the success of equilibrium concepts. We have formalized this definition as a stability concept—asymptotic stability—that is defined in relation to a hypothesis as to the behavior of evolution—the replicator dynamics. With this, we have established a framework in which we can assess the viability of an equilibrium concept—in terms of its agreement with the stability concept, and hence with our conditions for evolutionary significance.

1.5 Methodology and Assumptions of Limitative Results

The results gathered in this thesis collectively demonstrate that even under ideally favorable assumptions regarding the underlying dynamics and standards of stability—replicator dynamics and asymptotic stability—all candidate equilibrium concepts fall short. The approach taken here, and in much of the literature, is to test the equilibrium concepts under assumptions constituting a "best-case scenario." The idea is that, if the equilibrium solution concepts fail under such assumptions, a sort of informal "upper bounds" for the domain of their reliable application has been established, and so a strong case has been made against them. For these results to carry weight, we must justify why replicator dynamics and asymptotic stability constitute such a best case for our equilibrium solution concepts.

Among evolutionary dynamics, the replicator dynamics are commonly regarded as the simplest member of the class of deterministic dynamics. Our claim is that if an equilibrium concept fails to capture evolutionary significance for the replicator dynamics, then it has little hope to do so for other more complicated dynamics. The reason for this is that RD make largely the same assumptions about the population as the ESS concept does. This should not come as a surprise, as the RD were formulated by Taylor and Jonker [1973] specifically for the purpose of providing an explicit evolutionary dynamics for the ESS concept. And since all of our candidate equilibrium concepts are refinements of the Nash equilibrium concept, just as ESS is, they carry largely the same assumptions as to the nature of the evolutionary populations involved. Specifically, each of the equilibrium concepts implicitly assumes infinite population, no drift, no population structure, and a uniform environment [Taylor & Jonker, 1973]. It may be argued that the replicator-mutator dynamics (an augmentation of the replicator dynamics) better mirrors the assumptions in the ESS of the persistent threat of small mutation populations. However, we will consider in the discussion, how equilibrium

concepts are even less successful when mutation is explicitly introduced. The RD are, by design, uniquely congenial to our equilibrium concepts.

Similarly, asymptotic stability is the most common stability concept used in tandem with deterministic dynamics broadly, and the replicator dynamics specifically [Hofbauer, 1998; Cressman, 2003; Sandholm, 2009]. We will see that stronger stability conditions—those that pick out fewer (sets of) states as stable—will cause the equilibrium concepts to perform worse, as the equilibrium concepts are already too weak with respect to asymptotic stability, and pick out many states it deems as unstable. In the discussion, we consider how the equilibrium concepts perform under a weaker stability concept—Lyapunov stability. We will see that results indicate that they will performs even more poorly than under asymptotic stability. These results suggest that asymptotic stability may provide something in the neighborhood of locally optimal stability conditions for the replicator dynamics.

In short, the reasoning for choosing RD and AS as our assumptions under which to carry through our limitative results is that failure under these optimally favorable conditions constitutes a strong argument against equilibrium concepts, and shows that the domain under which they have any hope of providing a reliable form of analysis is small indeed.

2 The Limitations of Equilibrium Concepts Under the Replicator Dynamics

A natural place to begin the examination of the limitations of equilibrium concepts is at the Nash equilibrium (NE) concept. This is because, in their logical structure, all other equilibrium concepts are refinements of the NE concept. That is, each equilibrium concept can be thought of as beginning with the full set of Nash equilibria, and proceeding to eliminate those equilibria it deems are unviable. Unmodified, the NE is shown to be far too weak, and each refinement—strict NE, ESS, NSS—attempts to narrow in on the correct subset of equilibria that are to be all and only the evolutionarily significant states. The exception to this exclusive narrowing is the ESSet concept, as it also expands the range of equilibrium states to allow for sets of states that are collectively (though not necessarily individually) asymptotically stable. Each of these refinements will also be found to be, in one or more ways, too weak, in that they pick out (sets of) states that are not evolutionarily significant. Some equilibrium concepts are also too strong in that they fail to pick out evolutionarily significant (sets of) states. Additionally, we will find that all equilibrium concepts share a way in which they are uniformly too strong: They all fail to capture intuitively significant outcomes, such as limit cycles and strange attractors, that are not expressible in terms of sets of states.

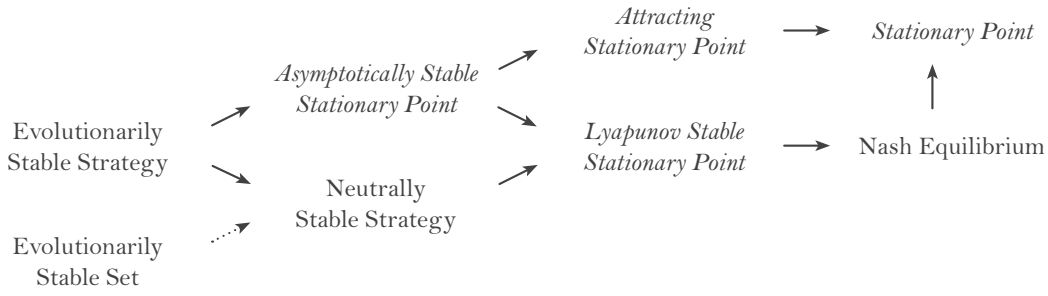


Figure 5: logical relations between equilibrium and stability concepts

Figure 5 depicts the flow of implication between equilibrium concepts. Note that all converse implications are not generally true. The dashed arrow from Evolutionarily Stable Set to Neutrally Stable Strategy is not implication, since non-singleton sets cannot be NSS. Rather, the dashed arrow represents the relation that if a state is *a member of* an ESSet, then it is an NSS.

2.1 Limits of the Nash Equilibrium

The Nash equilibrium is the fundamental equilibrium concept of classical, rational choice game theory. A state is a NE in which no individual stands to gain by unilaterally switching her strategy—i.e., by switching her strategy when all the strategies of all the other players remain fixed. For a population in such a state, no individual has positive incentive to deviate, and so it will be a stationary point. In other words, if there are no perturbations, a population starting at that state will remain at that state. Formally, we define the NE as follows:

> **Definition 2.1.1 (Nash Equilibrium)** A state $x^* \in X$ is a Nash equilibrium if, for all i such that $x_i \in \Delta$, $u(x_i^*, x_{-i}^*) \geq u(x_i, x_{-i}^*)$.

where $u(x_i, x_{-i}^*)$ is the payoff to an individual using the (mixed) strategy x_i given that all others players $-i$ are using their Nash equilibrium strategies in x_{-i}^*. For ease of explanation and understanding, but without loss of generality for the following results, we will formulate our definitions from here on for the case of symmetric pairwise interactions. The NE then becomes:

> **Definition 2.1.2 (Nash Equilibrium)** A state $x \in X$ is a *Nash equilibrium* if, for all $y \in \Delta$, $u(x, x) \geq u(y, x)$.

What is important to note about the Nash equilibrium solution concept is that all NE are stationary points under RD. The converse, however, is not true. Not all stationary points are NE. This is because, under RD, every unused strategy remains unused, and every pure strategy corresponds to a state of the population in which all but one strategy is unused. Hence, that single strategy will be the only strategy present for all future times, as its population proportion will remain constantly at one. Thus, every pure strategy is a stationary point of the dynamics. However, all interior stationary points are indeed NE [Skyrms, 1992].

The obvious shortcoming of the NE concept is that it can capture equilibria that are unstable and not attracting. Namely, it captures sources and saddles. Of the stationary points captured by the NE, there are three kinds: Sinks, sources, and saddles. Sources are, by definition, unstable, and cannot be attractors. Saddles may be either stable or unstable, and attracting or not attracting. Sinks are stable and attracting, though their basins of attraction may be extremely small.

To get an intuitive understanding of why it is problematic that NE may be sinks, we can imagine the 'fitness landscape' of an evolutionary game under RD. In such a landscape, as strategies with locally higher payoffs grow in proportion, the population will tend to 'ascend fitness slopes'. That is, given some initial

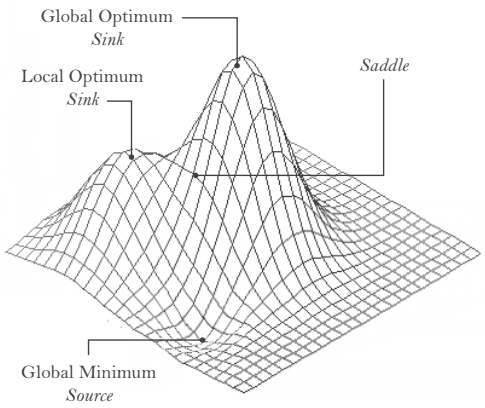


Figure 5.5: Fitness Landscape

conditions, strategies will grow as a monotonic function of their fitness, and so the population proportions of more successful strategies will grow larger. This will push the population toward the nearest accessible fitness peaks. That is, the population will move toward states where the pressure of fitter strategies reach a local maximum and balance one another out. If visualized in two or three dimensions, it will look like a kind of 'inverse gravity' is at play. In the fitness landscape, a sink is a local optimum—a convex peak—such that a population experiencing sufficiently small perturbations from this sate will gravitate back toward it. A Source is a local minimum—a concave bowl—such that a population experiencing any perturbation from this state will be repelled by it and continue in its trajectory away from this point. A saddle point is a stationary point that is neither a local optimum nor minimum. Along any single dimension it may be an inflection point, attracting from one side, and repelling from the other, or it may be perfectly uniform, with no slope on either sides. A population perturbed from a saddle may return, may be swept away, or may only move just as far as the distance it was perturbed.

	a	b
a	0	0
b	0	1

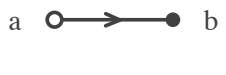


Figure 6: Unstable Nash Equilibrium

To demonstrate that the Nash equilibrium provides conditions too weak for evolutionary significance, we need only observe how sinks can be Nash equilibria. Consider the simple symmetric 2×2 game in *Figure* 6. Both the monomorphic population states—those composed entirely of strategies a or b, corresponding to the state vectors $\langle 1,0\rangle$ and $\langle 0,1\rangle$—are Nash equilibria. In each monomorphic state, no single individual stands to gain by unilaterally changing her strategy. Yet, the monomorphic population state $\langle 1,0\rangle$ composed of only a-types is not stable: It is a source, and the phase portrait reveals that the slightest perturbation will send the population straight off toward strategy b corresponding to the population state $\langle 0,1\rangle$.

Sources represent an extreme case of instability. No source can be asymptotically stable, and so, by our criteria, we can conclude that no source can be evolutionarily significant. Yet, NE may well be sources. This fact suffices to demonstrate that the Nash equilibrium is too weak a concept. This line of thinking provides the motivation for the following refinements of the NE concept

that strengthen its requirements with the aim of narrowing in on the subset of viable equilibria by eliminating implausible stationary points.

2.1.1 Limits of the Strict Nash Equilibrium

The obvious next-step, and refinement of the Nash equilibrium, is to strengthen it to the strict Nash equilibrium. The strict NE has the virtue of eliminating all the unstable states captured by the plain NE. Sinks cannot be strict NE. However, it is clearly too strong, and eliminates a great deal other than unstable NE. The strict NE eliminates crucial subsets of NE that can be evolutionarily significant, namely, the set of mixed strategy Nash equilibria.

To define the Nash equilibrium, we simply modify the NE by making the inequality between payoffs strict. Formally, we can define the strict Nash equilibrium for symmetric 2-player normal form games as follows

> **Definition 2.1.1.1 (Strict Nash Equilibrium)** A state $x \in X$ is a *strict Nash equilibrium* if, for all $y \in \Delta$ where $y \neq x$, $u(x,x) > u(y,x)$.

Why can't a mixed strategy NE be a strict NE? The argument goes as follows: without loss of generality, let a mixed strategy x be a NE. So we know that no player stands to gain by unilaterally changing strategies. But can this be a strict inequality? That is, must an individual stand to lose by unilaterally changing her strategies? Assume for the sake of contradiction that she does. This means that if any of the proportions of the pure strategies she is playing is changed, her payoff is made strictly less. And since x is nontrivially mixed, there is some pure strategy e_i in x that composes a proportion x_i of the population strictly greater than zero and strictly less than one. But then, if increasing (or decreasing) the value x_i lowers the payoff to the population, this implies that the payoff could be optimized by setting x_i to zero (or one). This contradicts the assumption that x is a mixed NE. Hence, the assumption is false, and no mixed strategy NE can be strict.

So, mixed strategies cannot be strict NE. But can we possibly abide the elimination of mixed strategies? We cannot, given that mixed strategy equilibria can be asymptotically stable, and so, by our standards, they can be evolutionary significant. In fact, there is a wealth of important examples in nature of what appear to be mixed strategy equilibria.

We consider an example of an asymptotically stable mixed strategy NE in the famous hawk-dove (HD) game. The HD game was introduced in tandem with the ESS concept by Maynard Smith [1973] to provide an explanation of ritualized animal conflict, and has since come to be a canonical model in evolutionary game theory.

Figures 7 and 8 display the game matrix and phase portrait for a simple HD game. In this game, we model a contest for resources between individuals who escalate conflict (Hawks, denoted by H), and individuals who do not (Doves, denoted by D). A Hawk, when competing with a Dove, wins all of the resource V under contention. But in conflict with another Hawk, both parties escalate, and so risk incurring a cost for injury C. Two doves, on the other hand, share the resource, or equivalently, have a half chance of acquiring it. If $C > \frac{V}{2}$, there are three Nash equilibria in this game: $\langle H, D \rangle$, $\langle D, H \rangle$, the polymorphic state where the proportion of the population using the Hawk strategy is $\frac{V}{2C}$ (e.g. ½ of the population uses each strategy when $V = C$).

	H	D
H	V/2−c	V
D	0	V/2

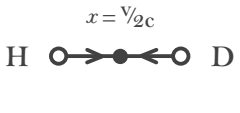


Figure 7: Hawk-Dove Class

	H	D
H	0	3
D	1	2

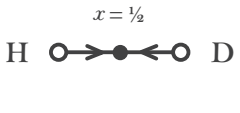


Figure 8: Hawk-Dove Game with V=C

As the phase portrait reveals, the polymorphic state represented by the mixed strategy equilibrium is indeed both stable and attracting. Any perturbation from the $\langle \frac{V}{2C}, 1 - \frac{V}{2C} \rangle$ stationary point will result in a return to that point. This is because an overabundance of any strategy creates an environment that yields a higher fitness for the complementary strategy. That is, more Hawks than the equilibrium proportion means that Hawks run into one another more often incurring more injuries, and into doves more seldom, acquiring fewer victories—their fitness will decrease more rapidly than that of the Doves', and the population will be brought back into balance. The dual of this occurs in the case of the overabundance of Doves; the Hawks have a heyday and so increase in proportion. Thus any perturbations from this mixed NE will result in a return to the equilibrium. Moreover, we can also see from the phase portrait that the population will converge to this equilibrium from any interior point. This is an asymptotically stable mixed strategy NE. There is an abundance of such interaction structures. Among these is the classes of anti-coordination games (of which HD is member) [Cressman, 2003].

Given the existence of asymptotically stable mixed NE, and the prevalence of interspecific variation in nature, the exclusion of mixed strategy equilibria provides more than enough justification for eschewing reliance on strict NE in our analysis of evolutionary games.

2.1.2 The Limits of Evolutionarily Stable Strategy

Introduced by Maynard Smith and Price in 1973, the evolutionarily stable strategy (ESS) is the flagship equilibrium concept of evolutionary games. When game theory texts introduce evolutionary game theory, they typically do so by presenting a primer on ESS [Osborne, 1994]. When ecologists employ game theoretic techniques to explain the persistence of a given trait in a population, they typically do so by identifying the ESS of the interaction. Yet, as we will see, the ESS has significant shortcomings. While a substantive improvement over normal and strict NE, the ESS concept still fails to capture many evolutionarily significant outcomes.

In a symmetric normal form game, an ESS is a strategy with the following property: A population in which nearly all individuals play this strategy is resistant to invasions by small groups of mutants playing any alternative strategy. Formally,

> **Definition 2.1.2.1 (Evolutionarily Stable Strategy)** a strategy x is *evolutionarily stable* if for every $y \neq x$ there exists an $\bar{\varepsilon}(y) > 0$ such that $(1-\varepsilon)\cdot u(x,x) + \varepsilon \cdot u(x,y) > (1-\varepsilon)\cdot u(y,x) + \varepsilon \cdot u(y,y)$ for $0 < \varepsilon < \bar{\varepsilon}(y)$.

In an ESS, the expected payoffs to the incumbent strategy x is strictly greater than that of any alternate strategy y, used by a sufficiently small ε proportion of mutants in the population. Equivalently, in a way that makes its relation to the NE clear, the ESS may be formulated in terms of the combination of two conditions:

> **Definition 2.1.2.2 (Evolutionarily Stable Strategy)** A strategy x is *evolutionarily stable* if for all $y \neq x$
> (1) $u(x,x) \geq u(y,x)$ for all $y \in X$, and
> (2) if $u(x,x) = u(y,x)$, then $u(x,y) > u(y,y)$.

The first condition says that the incumbent strategy x composes a (weak) NE with itself, and the second condition says that if there is a mutant y who fairs as well

against x as it does against itself, then x must do better against that mutant that it does against itself. That is, any mutant strategy must not be able to invade, and moreover must also be driven to extinction.

When compared to the strict NE, the ESS represents a clear move in the right direction. As with strict NE, with ESS we retain the guarantee of asymptotic stability, but unlike the strict NE, the ESS doesn't automatically disqualify mixed strategy equilibria.

In one direction, we have that if a state is an ESS, then it is indeed asymptotically stable. This guarantee is spoiled, however, by the fact that ESS can be set up to have arbitrarily small basins of attraction such that the criterion of attraction in asymptotical stability is technically satisfied, but the probabilities of starting at the initial conditions that would be delivered to such a state are practically negligible. Thus, we are confronted with the problem that there can be asymptotically stable states that are dissonant with our intuitive notion of evolutionarily significance.

Looking at the matrix and phase portrait in *Figure* 9, for any positive value of ε, no matter how small, the state $\langle 0,0,1\rangle$ is the only ESS of the game. Yet, the Lyapunov stable states composing the bottom edge can have as their basin of attraction nearly the entirety of the state space. None of these, either individually or collectively, constitute an ESS.

1	0	0
1	0	ε
0	0	2ε

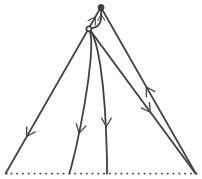


Figure 9: Small Basin of Attraction

1	1	1
0	0	3
2	1	0

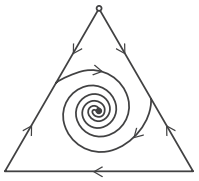


Figure 10: Globally Attracting Non-ES State

3	6	0
0	0	9
2	3	4

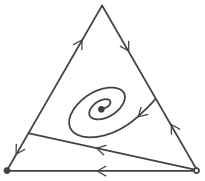


Figure 11: Zeeman's 1981 Attracting Non-ES State

In the other direction, the results are far worse. There are attracting states, and attracting sets of states that are not ESS. There are stable states and stable sets of states that are not ESS. And there are AS states, and collectively AS sets of states that are not ESS [Bomze, 1986, 1995]. We consider examples of non-ES states and sets of states, as these are both attracting and stable.

The reason that a stable attractor may not be an ESS is that the dynamics may spiral in elliptically towards the attractor. This cannot be covered by the notion of ESS due to the linearity inherent in its definition. To see this, we consider the

game in *Figure* 10, where we have a global attractor that is not and ESS. Here, the point $x = \langle \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \rangle$ is no ESS, but is globally attracting. It is readily observed that any interior state will converge to x, and we can confirm that x is not an ESS by working through Definition 2.1.2.2 of ESS. First we observe that if one individual uses x then the other player gets an expected payoff of 1 no matter what strategy they use. That is, for any alternate strategy y, $u(x, x) = 1 = u(x, y)$. So condition (1) of ESS is satisfied, and no strategy does better against x than it does against itself. But this payoff tie means that we need to check the condition (2), which is indeed not satisfied. To see this, consider a perturbed states such as $y = \langle 0, \frac{1}{2}, \frac{1}{2} \rangle$. Here, we get that $u(x, y) = 1 < 2 = u(y, y)$and so the mutant strategy y fairs better against itself than x does against it. But the mutant's advantage is temporary as y is unstable and gives way to the invasion of another mutant strain, which gives way to yet another, and so on, with an orbit that spirals in towards x. A similar example of an attractor with a large basin of attraction is provided by Zeeman's 1981 counterexample depicted in *Figure* 11.

What these examples show is that, from the point of view of RD, an attractor is a more general notion than an ESS, and a better characterization of resistance to mutation. Moreover, we have from Zeeman [1981] that if an ESS lies in the interior of the state space, then its basin of attraction is the whole interior, and so it is the only attractor. But if an attractor is in the interior of the state space, it may have a smaller basin of attraction, and the game may still admit other attractors on the boundary of the state space (see *Figure* 11).

A defining limitation of the ESS concept is that it captures only single population states, and so cannot capture non-singleton sets of states, whatever their basins attraction or stability properties; sets of states that are collectively asymptotically stable are beyond its reach. Consider the evolutionary game and dynamics in *Figure* 12. Examining the phase portrait, it is clear that interior trajectories will converge to the set $E = \{\langle a, b, c \rangle | \, a = b, c \in [0,1]\}$. But no member of E is an ESS. This is because every distinct strategy in this set is in a payoff tie against any other, and so the second condition of the ESS, that of strict inequality, will not be satisfied. Yet, when considered collectively, this set is both globally attracting and stable—we will see that it is an ESSet.

0	2	0
2	0	0
1	1	0

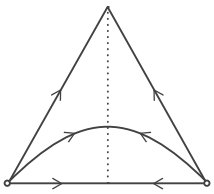


Figure 12: Asymptotically Stable Set

Why should we worry about sets of states *qua* sets? In normal form games, sets of stationary points tend not to be robust under small perturbations in the payoff

structure of the game. Sets containing continua of equilibria can often be made to collapse to a single state. In extensive form games, however, this is not so. It is a well-known property of extensive form game with nontrivial structure that they produce non-singleton Nash sets [Cressman, 2003]. Importantly, it has been shown that, in many extensive form games, these sets are indeed robust under perturbations of payoffs [Huttegger, 2009]. Moreover, as in *Figure* 12, no single state in the attracting set is evolutionarily significant, since the set of initial conditions that will converge to any point in E is a measure zero set. It is only the set as a whole that commands convergence from a non-measure-zero set of initial conditions, and so only the set as a whole qualifies as evolutionarily significant. Sets of states that are significant only as sets should be expected, particularly in extensive form games, and should be accounted for.

Before moving on, it may be interesting to consider how ESS fairs in important special classes of games cases. In 2×2 symmetric games we have the strong result that all asymptotically stable states are ESS, and so here their relationship becomes one of biconditional implication [Cressman, 2003]. In 3×3 symmetric games we find the lowest level of complexity of game structure that gives rise to stable and attracting states that are not ESS, as well as limit cycles, and other complex disequilibria behavior [Bomze, 1985]. And in 4×4 games, we have the debut of strange attractors and chaotic behavior [Skyrms, 1991, 1992].

1.05	0	0	1.5
2.05	0.6	0	0
0	0.4	0	2.25
1.55	0.5	0.1	0.5

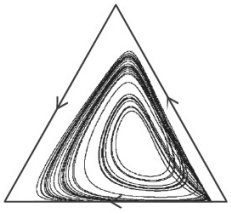


Figure 13: A Strange Attractor

Can we hope for better results in games with more strategies? Might there be some windfall such that 7×11 games somehow turn out to be reliably assessed by equilibrium concepts? We know this is cannot be the case, because we can imbed smaller games into larger ones [Cressman, 2003]. So, if we show that ESS fails to capture asymptotically stable elliptical attractors in 3×3 games, then we know that this holds for all $m \times n$ games where m and n are both greater than or equal to 3. Thus, the results we have provided generalize to all classes of more complicated games.

In summary, the conditions provided by the ESS concept are too weak in that they admit asymptotically stable states with potentially negligible basins of attraction. They are also clearly too strong in that they fail to capture (globally) attracting states, as well as some asymptotically stable states, and all collectively asymptotically stable sets of states.

2.1.3 The Limits of the Neutrally Stable Strategy

The neutrally stable strategy (NSS) relaxes the ESS concept to weak inequality so as to include more evolutionarily significant equilibria. Particularly, it redeems some, but not all of the Lyapunov stable state excluded by the ESS. An advantage of the NSS over the ESS is that it can account for members of significant Nash components, as well as portions of some limit cycles. In the latter case, the hubs of limit cycles can be NSS [e.g. in the RPS game], why the orbits themselves will go uncaptured. A basic property of the NSS under RD is that it is Lyapunov stable [Bomze and Weibull, 1995]. The converse is not true, and we will see that there can be Lyapunov stable states that are not NSS. A NSS is not necessarily attracting, and so it is not necessarily asymptotically stable. There are evolutionarily insignificant NSS, as demonstrated by the $n > 1$-player degenerate game $A = [0]$, where every point in the state space is a NSS with a measure-zero basin of attraction. The NSS captures some evolutionarily significant states neglected by the ESS at the expense of letting other insignificant states in.

Intuitively, at a NSS, mutant strategies will not be able to invade, but neither will they be driven out. Formally, the NSS is a weakening of the ESS to non-strict inequality in condition (2).

> **Definition 2.1.3.1 (Neutrally Stable Strategy)** A strategy x is *neutrally stable* if for all $y \neq x$
> $(1)\ u(x,x) \geq u(y,x)$ for all $y \in X$, and
> (2) if $u(x,x) = u(y,x)$, then $u(x,y) \geq u(y,y)$.

To see what significant states the NSS captures that the ESS does not, we revisit the game in $Figure\ 12$ with an asymptotically stable set, none of the members of which are ESS. Because of the relaxing of condition (2) to non-strict inequality in the NSS, all of the states that are eliminated by the ESS remain; the NSS captures each of the states belonging to the set. Again, however, NSS is defined for individual states and cannot account for the set *qua* set. This is unfortunate, since it is the set which is asymptotically stable, whereas any one of its members is only Lyapunov stable.

Let us consider an example demonstrating that the converse to the essential property of NSS does not hold—that not all Lyapunov stable

1	1	0
2	0	0
0	2	0

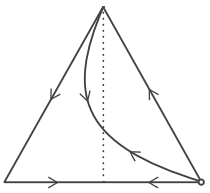


Figure 14: Lyapunov Stable Non-NS States

states are NSS. Consider the game and state space in *Figure* 14. Here, all population states $x = \langle a, b, c\rangle$ with $a = b > 0$ are Lyapunov stable, but only those with $1/3 \leq a \leq 1/2$ are also NS. Define the set of Laypunov stable states as $\Delta^{LS} = \{\langle a, b, c\rangle | a = b > 0, c \in [0,1]\}$ and the set of NSS $\Delta^{NS} = \{(a, b, c) | a = b > 0, a \in \left[\frac{1}{3}, \frac{1}{2}\right], c \in [0,1]\}$. Let x belong to Δ^{LS}, say $x = \langle\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\rangle$. Then $u(x, x) = \frac{1}{4}$, and x is a NE with itself, however, it is not the case that $u(x, y) \geq u(y, y)$ for all y. Consider $y = \langle 1,0,0\rangle$, then $u(y, y) = 1 > \frac{1}{4} = u(x, y)$. Thus x is Lyapunov stable but no NSS.

2.1.4 The Limits of Evolutionarily Stable Sets

The Evolutionarily stable set (ESSets) was introduced by Bernard Thomas in 1985 and is the set-valued generalization of the ESS. In many senses, ESSets represent the best option on the table. The ESSet contains the ESS as a special case—the singleton ESSet—and so it captures all of the evolutionarily significant that the ESS does. On top of this, all non-singleton ESSets are also asymptotically stable, and so it captures evolutionarily significant outcomes excluded by the ESS. ESSet can be global attractors in a broader range of cases than the ESS. Specifically, if an ESSet set contains an interior NE, then it is globally asymptotically stable [Cressman, 2003]. Moreover, by construction, the ESSet fairs no worse than the ESS in capturing evolutionarily non-significant outcomes.

All members of an ESSet are NSS [Weibull, 1995], though there are NSS that will not belong to ESSets [Bomze, 1985]. From this it follows directly that there are Lyapunov stable states that are not in ESSets. But there are also non-NSS Lyapunov stable states that do not belong to an ESSet. Unfortunately, many of the asymptotically stable states that are not ESS are also not ESSets. And, as with ESS, ESSets can exhibit extremely small basins of attraction [Cressman, 2003].

> **Definition 2.1.4.1 (Evolutionarily Stable Set)** A closed non-empty set of strategies S is an *ESSet* if and only if for each $x \in S$ there exists a neighborhood B around x such that for all $y \in B$,
> (1) If $y \in S$, then $u(x, y) \geq u(y, y)$; and
> (2) If $y \notin S$, then $u(x, y) > u(y, y)$.

Now, we can account for the central outcome of the games like that in *Figure* 12. Recall the set $S = \{\langle a, b, c\rangle > | \, a = b, c \in [0,1]\}$ could not be captured by the ESS, and was only captured in terms of its members—which are

not individually asymptotically stable—by the NSS. But it is an ESSet. To see this, let $x \in S$ and $y \notin S$. So x and y are of the form $\langle a, b, c\rangle$ and where $a = b$ and $a \neq b$ respectively. By looking at the game matrix, we see that by fixing the value of c, the game reduces to the upper left $2x2$ subgame where the only symmetric Nash equilibrium is where $a = b$. Thus for any such $y \notin S$, there is an $x \in S$ where x such that $u(x, y) > u(y, y)$ as required by (1). On the other hand, if $y \in S$, by simple algebra it can be shown that that $u(x, y) = \mathrm{u}(\mathrm{y}, \mathrm{y})$.

The games in *Figures* 10 & 11 —the globally attracting non-ES state, and Zeeman's 1981 counterexample—both also work as counterexample for the ESSet. This follows from the fact that they are both singleton sets of non-ES equilibria. Nothing in the ESSet concept is added that can deal with them, and the same arguments provided in the case of ESS carry through. Thus we still have asymptotically stable states that are not ESSets.

Moreover, we can find evolutionarily significant sets that are not ESSets. In *Figure* 15 we find an examples of attracting sets of NSS that do not belong to an ESSet. To see how this is so, consider the set $E = \{\langle x, y, 0\rangle \mid y \geq \frac{1}{2}\}$. This is not an ESSet, because of the boundary point $x =< \frac{1}{2}, \frac{1}{2}, 0 >$. This can be observed in how the state $y = \langle \frac{1}{2} - \varepsilon, \frac{1}{2} - \varepsilon, 2\varepsilon\rangle$ fares. We find that the payoff to x against y is $u(x, y) = -\varepsilon$, while the payoff to y against itself is $u(y, y) = -\frac{\varepsilon}{2} - \varepsilon^2$. For sufficiently small values of ε we get that $u(y, y) > u(x, y)$ violating condition (2), and making it so that E is no ESSet. Because a single boundary point of the set is not attracting, the set as a whole is technically not attracting. Yet we can anticipate that all interior trajectories will end up at the set of stable states comprising E. The set E then constitutes an evolutionarily significant outcome by our definition, but is not captured by the ESSet concept.

0	1	0
0	1	1
1	0	1

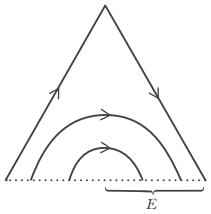


Figure 15: NSS Not Belonging to an ESSet

2.2 Summary and Discussion

In summary, we have seen that—for the replicator dynamics—the original Nash equilibrium concept is far too weak. The NE captures all interior stationary points, but among these we find many unstable states (*Figure* 6). Its natural modification, the strict Nash equilibrium, is far too strong. Strict NE eliminates the unstable

states captured by the NE, but fails to capture asymptotically stable mixed strategies (*Figure* 8). The evolutionarily stable strategy is also too strong. ESS captures many of the AS mixed strategies excluded by the strict NE, but fails to capture AS states where the dynamic flows elliptically towards the attractor (*Figures* 10, 11). The neutrally stable strategy is somewhat more inclusive than the ESS, but suffers from the same central limitations (*Figure* 14). Each of the previous concepts is also too strong in that they fail to capture sets of states that are collectively (though not necessarily individually) AS (*Figure* 12). The most sophisticated equilibrium concept, the evolutionarily stable set, does capture some sets of states that are collectively asymptotically stable, though it fails to captures others (*Figure* 15), and falls prey to elliptically attracting AS states (*Figures* 10, 11) just as the ESS does. When evaluated against asymptotic stability, each of these concepts comes up short. In this comparison, the ESSet, ESS, and strict NE turn out to be too strong, while the NSS and NE turn out to be both too strong and too weak.

Table 1: Equilibrium Concepts *for* Replicator Dynamics

	Too Weak Captures some outcomes that ARE NOT evolutionarily significant that are:			**Too Strong** Fails to capture some outcomes that ARE evolutionarily significant that are:		
	Unstable	Not Attracting	Small Basins of Attraction	Sets of States (as Sets)	Asymptotically Stable States	Cycles
Nash Equilibrium (NE)	☑ *Figure 6*	☑ *Figure 6*	☑ *Figure 9*	☑ *Figure 12*	☐	☑ *Figure 3*
Strict Nash Equilibrium (Strict NE)	☐	☐	☑ *Figure 9*	☑ *Figure 12*	☑ *Figures 8, 10, 11*	☑ *Figure 3*
Evolutionarily Stable Strategy (ESS)	☐	☐	☑ *Figure 9*	☑ *Figure 12*	☑ *Figures 10, 11*	☑ *Figure 3*
Neutrally Stable State (NSS)	☐	☑ *Figures 6, 14*	☑ *Figure 9*	☑ *Figure 12*	☑ *Figures 10, 11*	☑ *Figure 3*
Evolutionarily Stable Set (ESSet)	☐	☐	☑ *Figure 9*	☐	☑ *Figures 10, 11*	☑ *Figure 3*

Over and above these shortcomings, there are two additional limitations common to all the equilibrium concepts. First, every equilibrium concept can be seen as too weak in the shared sense that they can pick out states with basins of attraction that can be made arbitrarily small (*Figure* 9). While extremely small basins of attraction technically satisfy the conditions of asymptotic stability, they are at odds with our informal notion of evolutionary significance. That is, they do not seem to constitute probable outcomes of an evolutionary process. The remedy for this is not obvious, as a general stipulation of some lower bound on basin size seems *prima facie* unprincipled and unsatisfactory.

Second, there is also a distinct in-principle argument for the claim that the equilibrium concepts are all too strong. The argument is as follows: Equilibrium concepts are formulated exclusively in terms of states or sets of states. But there are outcomes that cannot be effectively expressed in these terms, for example limit cycles (*Figure* 3), and strange attractors (*Figure* 13). While not asymptotically stable, these are dependable regularities that can command convergence from a significant set of initial conditions of a system, hence we are compelled to say that they constitute evolutionarily significant outcomes. But then there exist evolutionarily significant outcome that the equilibrium concepts fail to capture. Indeed, on this count, equilibrium concepts are formulated in the wrong way to begin with. That is, this problem has no hope of being resolved by strengthening or weakening our concepts in just the right way. Any solution would necessarily involve reformulating the equilibrium concepts in very different terms. How to go about this is not apparent. In fact, the problem of negligible basins of attraction is not unique to equilibrium concepts—stability concepts too are formulated in terms of (sets of) states, and so are vulnerable in the same way.

The challenge is that behavior, such as limit cycles and strange attractors seem to be best described in differing qualitative terms. Closed cycles are often described by providing the differential equations that detail their orbits [Cressman, 2003], while describing the behavior of strange attractors, or even inferring their existence, can demand the use of numerical methods [Skyrms, 1992]. Given this, it may be unreasonable to expect the equilibrium or stability concepts to provide such descriptions in addition to their point- and set-valued predictions. It is reasonable, however, to take the presence of such evolutionarily significant disequilibria behavior as one more reason to avoid exclusive reliance on equilibrium concepts, and to engage in practice of explicitly examining nontrivial dynamics on a game-by-game basis.

2.2.1 Results Under Different Assumptions: Lyapunov Stability

We have seen that the equilibrium concepts fail to reliably capture evolutionary significance when our model of evolution is the replicator dynamics, and our stability concept is asymptotic stability. We provided an argument in §3 for why RD and AS constitute uniquely favorable assumptions under which to assess our equilibrium concepts. However, RD & AS by no means exhaust the possibility space of plausible assumptions. It may be informative to consider alternatives to each. First, we consider what happens to our results when we weaken the stability condition to Lyapunov stability, and in §3 we consider what occurs when we take stochastic dynamics as our model of evolution.

Recall that Lyapunov stability is the weakening of asymptotic stability by withdrawing the demand that a stable state also be attracting (see *Definition* 1.3.1). It follows that all asymptotically stable states are Lyapunov stable, but the converse is not true. We note also that Lyapunov stability does not align well with the definition of evolutionary significance that we have developed here. This is because, in giving up the requirement for attraction from a significant set of initial conditions, we include what we have called improbable outcomes—ones that can be arrived at only by starting at measure-zero subsets of the state space.

Revisiting our results with Lyapunov stability as the stability concept by which to evaluate the success and failure of the equilibrium concepts, we find that every equilibrium concept, except possibly for the NSS, fairs worse. The reason for this is that all of the asymptotically stable states not captured by the equilibrium concepts are also Lyapunov stable states. But then, there are also more Lyapunov stable states that are not asymptotically stable and that are not captured by any equilibrium concept [Bomze, 1986, 1995]. Hence the set of asymptotically stable states not captured by the equilibrium concepts is a strict subset of the set of Lyapunov stable states not captured by them. Even the NSS, which was designed specifically to try to retrieve stable states excluded by the ESS, fails to capture all Lyapunov stable states (*Figure* 14). In short, when we evaluate our equilibrium concepts against Lyapunov stability, the results are actually worse.

3 The Limitations of Equilibrium Concepts Under the Moran Process

3.1 Stochastic Models of Evolution

Stochastic models of evolution represent an important context in which to assess the success of equilibrium concepts that purport to predict the probable outcomes of evolution. This is because stochastic dynamics provide more realistic descriptions of evolutionary processes. Stochastic models allow us to assume finite population sizes, and to account for important evolutionary behavior such as drift [Taylor et al, 2004]. Indeed, we often buttress our faith in deterministic models of evolution by noting their conditional agreement with stochastic ones [Sandholm, 2009; Nowak, 2006; Hofbauer, 1998]. While our equilibrium concepts fail to reliably capture evolutionary significance in the context of deterministic dynamics, they would still be vindicated by capturing it in the context of stochastic dynamics. But as we shall see, however, stochastic dynamics paint a more complex picture of evolution that is less amenable to equilibrium analysis. Stochastic dynamics display a greater variety of qualitative behaviors that depend crucially upon population size, frequency of mutation, and intensity of selection.

To assess the performance of equilibrium concepts for stochastic dynamics, we must revisit to our new definition of evolutionary significance to guide us in selecting success criteria for the stochastic context. There has been a great deal of work analyzing stochastic evolutionary dynamics [Young 1998; Taylor et al, 2004; Fudenberg et al, 2006; Nowak, 2006; Sandholm, 2009], and so there have been many proposals for how best to characterize their behavior, and predict their probable outcomes. Prime among these proposals are those of the ESS^N, proposed by Nowak [2006], which attempts to translate the concept behind the ESS into stochastic terms via fixation probabilities, and of *stochastic stability*, introduced to game theory by Young [1993; 1998], which aims to picks out the most probable outcome of a stochastic process via stationary distributions. These will be discussed as we select formalizations of evolutionary significance for the Moran process. The stability concepts will enable us to assess the performance of the equilibrium concepts for our stochastic dynamics, and to contrast our results with those already obtained in the deterministic context.

3.1.1 The Moran Process

The standard stochastic model of evolution is the Moran Process [Moran, 1962], The Moran process is a Markov process in which, for each time step, two individuals are chosen: one for reproduction and the other for elimination. The process is defined over the state space χ by the designation of transition probabilities given by a transition matrix P between adjacent states. The leading idea in the shift from deterministic dynamics to stochastic dynamics is that processes beginning in identical states need not follow identical trajectories. Indeed, stochastic processes with identical initial conditions will typically go on to trace distinct paths through the state space. This means that the phase portraits we have thus far used to illustrate the flow of the replicator dynamics are no longer appropriate. We need different analytic tools to characterize the behavior and stability properties of stochastic evolutionary processes.

We may begin by considering a population of N individuals playing a 2x2 symmetric game with the payoff matrix

$$\begin{matrix} & A & B \\ A & a & b \\ B & c & d \end{matrix}$$

Where a, b, c, d are strictly positive. The state of the system is denoted by the number i of A-type individuals so that $N - i$ is the number of B-type individuals. The expected payoff for interaction to each type is given by

$$U_A = ax_A + bx_B$$
$$U_B = cx_A + dx_B$$

where x_A, x_B denote the proportions of A and B-types respectively. Given random pairwise interactions, the fitness of each strategy A and B as a function of the number of A-types i is given by

$$f_i = 1 - w + w\frac{a(i-1) + b(N-i)}{N-1}$$
$$g_i = 1 - w + w\frac{ci + d(N-i-1)}{N-1}$$

where w is the intensity of selection, so $w = 1$ implies that the individuals' fitness is entirely determined by their interactions in this game, and $w = 0$ implies that the game makes no contribution to the fitness of individuals.

Observe that we are accounting for the fact that individuals do not play against themselves. For the replicator dynamics this was a moot point since a single individual made no difference in an infinite population. Here, this fact means that individuals are slightly less likely to interact with those of their type (i.e. there is "negative correlation" [Skyrms, 2007]), and this and accounts for the advantage enjoyed by "spiteful" strategies [Hamilton, 1978] in small finite populations where the effect is more pronounced. As expected, the benefit to spiteful strategies is washed out, as the population grows large.

Individuals reproduce at a rate proportional to their fitness. The rate of reproduction then for A types is if_i and for B-types is $(\mathrm{N}-\mathrm{i})g_i$. At each period one offspring is chosen at random to enter the population. So, the probability of adding an A offspring is $\frac{if_i}{if_i+(\mathrm{N}-\mathrm{i})g_i}$, and the probability of adding a B offspring is $\frac{(N-i)g_i}{if_i+(\mathrm{N}-\mathrm{i})g_i}$. After reproduction, one individual is chosen at random to be removed from the population, so that with probability $\frac{i}{N}$ an A-type is removed, and with probability $\frac{N-i}{N}$ a B-type is removed. This makes it so that the population size remains constant.

We define the Moran process with population size N formally as a Markov Process $\{X_t^N: t \geq 0\}$ over the finite state space $\chi^N = \{x \in X: Nx \in \mathbb{Z}^N\}$ with transition probabilities given by

$$P_{i,j} = \begin{cases} \frac{N-i}{N}\frac{if_i}{if_i+(\mathrm{N}-\mathrm{i})g_i}, & \text{if } j = i+1 \\ \frac{i}{N}\frac{(N-i)g_i}{if_i+(\mathrm{N}-\mathrm{i})g_i}, & \text{if } j = i-1 \\ 1-P_{i,i+1}-P_{i,i-1}, & \text{if } j = i \\ 0, & \text{otherwise} \end{cases}$$

which composes a tri-diagonal matrix. We note that $P_{0,0} = P_{N,N} = 1$, so that the process has two absorbing states, $i = 0$ and $i = N$, and that all other states are transient. An *absorbing state* is a state that, once visited by the process, is never left. A *transient state* is one that can only be visited a finite number of times before the process is absorbed. If the population reaches either absorbing state, it will stay there forever, and as time t goes to infinite, with probability one, the process will reach one or the other absorbing state. This behavior is key to understanding the Moran process, and is captured by the "neutral drift principle": If a population contains several different types, then—in the absence of mutation—all but one type will become extinct [Nowak, 2006]. This is the case even if all the types

have the same fitness. From this it follows that no polymorphic state can be an outcome of the process, in the long run.

In any game, there will be as many recurrent classes (composed of singleton absorbing states) as the cardinality of the set of strategies $|S|$, and there will be $\sum_{k=2}^{|S|}\binom{|S|}{k}$ transient classes, where classes are defined as maximal sets of states that are mutually reachable from one another. For a state to be *reachable* from another means that there exists a time T such that the probability of arriving at the former given that we started at the latter is nonzero.

As a Markov process, the Moran process employs the assumption that future states of the process are probabilistically independent of past states given the present state. Formally, $Pr(X_{t+s} = j|X_t = i, X_u = k) = Pr(X_{t+s} = j|X_t = i)$ for all $i, j, k \in X^N$, $s \geq 0$, and $t > u \geq 0$. Additionally, because individuals in our model switch strategies sequentially, transitions of the process $\{X_t^N\}$ are always between adjacent states. Since the states are also linearly ordered, the Moran process falls into the class of Markov processes known as birth and death processes.

For now, we observe that in its simplest form, the Moran process assumes: random pairwise interaction, the absence of spatial or social structure, asexual reproduction, a constant environment, constant population size, perfect transmission of strategies from one generation to the next, and a corresponding absence of mutation. The latter two assumptions will be relaxed when we consider the Moran process with mutation.

Method of Analysis

How are we to parse and analyze the behavior of the Moran process? This is typically done by examining the probabilities that the process will end up in any of the monomorphic absorbing states when starting from given initial conditions. For our Moran process $\{X_t^N\}$, let ϕ_i denote the probability of the process ending up in state N when starting from state i. Then we have the recursive relation

$$\phi_i = P_{i,i+1}\phi_{i+1} + P_{i,i}\phi_i + P_{i,i-1}\phi_{i-1}$$

with boundary conditions $\phi_N = 1$ and $\phi_0 = 0$. The solution to this is given by Karlin and Taylor [1975]

$$\phi_i = \frac{1 + \sum_{j=1}^{i-1}\prod_{k=1}^{j}\frac{g_k}{f_k}}{1 + \sum_{j=1}^{N-1}\prod_{k=1}^{j}\frac{g_k}{f_k}}$$

In our analysis, we will be interested in the probability of a single mutant coming to replace the incumbent strategies, and take over the entire population. For our model, this is the probability that a process starting from a single A individual reaches the state composed of all A individuals. This is the called the *fixation probability* of A, and is given by

$$\rho_{AB} = \phi_1 = \frac{1}{1+\sum_{j=1}^{N-1}\prod_{k=1}^{j}\frac{g_k}{f_k}}$$

Conversely, the probability that the process starting from a single B individual reaches the all-B state is given by

$$\rho_{BA} = 1 - \phi_{N-1} = \frac{1}{1+\sum_{j=1}^{N-1}\prod_{k=1}^{N-1}\frac{f_k}{g_k}}$$

We can observe that

$$\frac{\rho_{AB}}{\rho_{BA}} = \prod_{k=1}^{N-1}\frac{f_k}{g_k}.$$

where if $\frac{\rho_{AB}}{\rho_{BA}}>1$, then it is more likely that a single A mutant comes to fixation in the population composed of $N-1$ B-types than the other way around.

Fixation probabilities of strategies can be compared to those of a neutral mutant (where $a = c = b = d$), which will reach fixation with a probability of $1/N$. A mutant with relative fitness r (where $a = c, b = d$ and $ar = b$) will have a fixation probability $\rho = \frac{1-1/r}{1-1/r^N}$.

Typically, fixation probabilities are difficult to compute for anything but the smallest populations, or when we take the weak selection limit $w \to 0$. In particular, for the realistic (and hence, important) case of large but finite populations, analytical results are often intractable. In these cases we can accurately estimate probabilities that a process ends up at any state given a distribution over its initial conditions using the Chapman-Kolmogorov (CK) equation given by

$$P^t = (P)^t$$

which says that the n^{th}-step transition matrix for the process is equal to the first-step transition matrix raised to the n^{th} power [Strook, 2005]. We will employ the CK equation to find the limit distributions of Markov processes, particularly when we deal with realistic parameterizations of both the Moran process and Moran process with mutation.

Before we proceed, there is a key connection between the Moran process and replicator dynamics we should discuss. For large population size, the replicator

dynamics describes the mean field (expected motion) of the Moran process. It can be demonstrated that as $N \to \infty$, the Moran process will, with high probability, remain close to the path of the replicator dynamics for some large but finite time T [Weibull, 2003]. This congruence of the behavior of the two dynamics for long but finite durations can be readily observed in simulations. For example, in the RPS game with cycles it can be seen that the process behaves as the replicator dynamics indicates for a significant duration before being absorbed into one of the monomorphic states.

The focus on determinist equations then, can be justified by an informal appeal to the law of large numbers. Since all randomness in the evolutionary model is idiosyncratic, it should be averaged in the aggregate as long the population size is sufficiently large. This mean-field relationship between the replicator dynamics and the Moran process is made precise in Sandholm [2009]. But if we want to understand the behavior of evolution for finite populations, in the presence of non-vanishing mutation, or for infinite time horizons, we must study the Markov process itself.

3.1.2 The Moran Process with Mutation

With the introduction of mutation, the Moran process changes qualitatively both in its behavior and in the tools required to analyze its behavior. Importantly, the process becomes irreducible. That is, there is a positive probability that the new perturbed process $\{X_t^{N,\eta}\}$ with mutation rate η, will transit from any given state i to any other state j within a finite number of periods. Thus every state is reachable by any other, and there is now only a single (recursive) class of states composed of the entire state space. Since absorption into a single state will not occur, fixation probabilities are no longer appropriate, and a different way of describing the behavior of the dynamics is needed.

We define the Moran process with mutation with finite population N and mutation frequency η formally as an irreducible Markov process $\{X_t^{N,\eta}: t \geq 0\}$ over the finite state space $\chi^N = \{x \in X: Nx \in \mathbb{Z}^N\}$ with transition probabilities given by

$$\hat{P}_{i,j} = \begin{cases} (1-\eta)\dfrac{N-i}{N}\dfrac{if_i}{if_i + (\mathrm{N}-\mathrm{i})g_i} + \eta\dfrac{N-i}{N}\dfrac{(N-i)g_i}{if_i + (\mathrm{N}-\mathrm{i})g_i}, & \text{if } j = i+1 \\ (1-\eta)\dfrac{i}{N}\dfrac{(N-i)g_i}{if_i + (\mathrm{N}-\mathrm{i})g_i} + \eta\dfrac{i}{N}\dfrac{if_i}{if_i + (\mathrm{N}-\mathrm{i})g_i}, & \text{if } j = i-1 \\ 1 - \hat{P}_{i,i+1} - \hat{P}_{i,i-1}, & \text{if } j = i \\ 0, & \text{otherwise} \end{cases}$$

where $\hat{P}_{0,0} = \hat{P}_{N,N} = 1 - \eta$. Note the structure of the mutation terms. What they say is that, most of the time $(1-\eta)$, selection behaves as normal, but in a minority of instances η, an offspring that was to be an A-type will become a B-type, or an offspring that was to be a B-type will end up as an A-type. Here, we have assumed that mutation is symmetric, but it need not be so. Asymmetric mutation can be accounted for by formulating the rate of mutation for one type a ratio of the other $r\eta$ where r is a positive constant.

The process $\{X_t^{N,\eta}\}$ typically moves in the direction indicated by the replicator dynamics. If the process begins in the basin of attraction of an AS state or other attractor of RD, then the initial period of evolution typically results in convergence to and lingering near this locally stable set. However, since the process $\{X_t^{N,\eta}\}$ is irreducible this cannot be the end of the story. Indeed, the process reaches all states in χ^N, and visits them infinitely many times in the limit of time. This means that the process must eventually leave the basin of the stable state and to enter the basin of a new attractor. The process continues in this way with long periods near each attractor broken up by sudden transitions between stable state. Which states are visited more often in the long run is determined by the relative improbability of these rare transitions [Young, 1998].

Method of Analysis

The traditional mode of analysis of the Moran process with mutation (MPM) is to relate the long-run behavior of the process with that of the simple Moran process without mutation (MP) by taking the limit of vanishing mutations. We saw, in the MP, that all an only monomorphic states were absorbing states. Now, the presence of mutations leads to rare fluctuation between these monomorphic states interspersed by long stays at each. Analysis can determine the proportion of time spent at each state in the limit of time. To do this, we find the stationary (or limit) distribution for the process as the frequency of mutation η approaches zero. Formally, we define a stationary distribution as follows

3.1.2.1 Definition (Stationary Distribution): A probability distribution $\mu \in \mathbb{R}^X$ is a *stationary distribution* of the irreducible Markov process $\{X_t^{N,\eta}\}$ if

$$\sum_{i \in X} \mu_i P_{i,j} = \mu \text{ for all } j \in X.$$

That is, a stationary distribution is a probability vector such that taking the product of the stationary distribution and the matrix of transition probabilities simply returns the stationary distribution. If we use the notation $Pr_\pi(\cdot)$ to mean $\{X_t^{N,\eta}\}$ is being run from the initial distribution π, then

$$Pr_\mu(X_t = i) = \mu_i \text{ for all } i \in X, t \geq 0$$

In other words, if a process starts off in its stationary distribution, it remains in this distribution at all subsequent times t. But what happens to the process if it starts in an arbitrary initial distribution π? As t grows large, the distribution of time spent by the process $\{X_t^{N,\eta}\}$ at each state converges to μ. Thus, from the ex ante point of view, the probable locations of the process $\{X_t^{N,\eta}\}$ over sufficiently long periods of time is characterized by μ.

The stationary distribution captures infinite-horizon behavior of the process in two distinct senses: It is both the process's limiting distribution and its limiting empirical distribution. That is, it captures both where we may expect to locate the process in the limit of time, and the proportion of time it will spend at each of the states in the state space.

The fact that the process is irreducible implies that it admits a unique stationary distribution [Strook, 2005]. We may also calculate stationary distributions for the unperturbed Moran process $\{X_t^N\}$, but they will not be unique. Rather, stationary distributions of the Moran process without mutation are sensitive to initial conditions. Consider the case when the process starts at an absorbing state: It will remain there forever. Hence, stationary distributions are not generally informative in the same way for the Moran process—specifically, they do not tell us the probable outcomes of the process $\{X_t^N\}$ unless we already know its initial conditions.

In general, computing the stationary distribution of the MPM $\{X_t^{N,\eta}\}$ means finding the (right) eigenvector of a matrix, a task that is computationally daunting unless the state space, and hence dimension of the matrix, is small. In these cases, we turn to the CK equations for a computational approximation at large values for time T.

For the Moran process $\{X_t^{N,\eta}\}$, in the limit of time, large population, and vanishing mutation, the stationary distribution collapses (typically) into a single point mass around one of the stable states. If the replicator dynamic has a unique asymptotically stable state, then the stationary distribution will concentrate around this state. But if there is more than one AS state, the stationary distribution will typically concentrate around only one of them. Only in rare cases, where the selection forces are perfectly balanced between multiple AS states, will stationary distribution spread its mass between more than one state [Fudenberg et al, 2006].

Having said that, it is important that these collapses occur only when we take either the small mutation limit for infinite horizon play. Such collapses do not generally characterize the behavior of evolution over long but finite durations and large populations, even when mutation is quite rare. This indicates something important about our limit methodology. As we will see, the behavior of the MP changes dramatically when we move away from limit assumptions. When modeling evolutionary processes, it can be important to assume population sizes and mutation rates commensurate with what we expect to occur in the target systems we are modeling.

Moreover, in order to place credence in the history-independent predictions provided by the stationary distribution, the relevant time horizon must be sufficiently long to justify appeal to this distribution. One can gauge the amount of time needed before the stationary distribution becomes relevant by determining how quickly the distribution of the evolutionary process approaches the stationary distribution (See Sandholm [2009, Ch.11] for a presentation of the relevant techniques).

3.2 Evolutionary Significance in Stochastic Dynamics

The deterministic replicator dynamics and stochastic Moran process are formulated in quite different mathematical languages. The former is specified by a system of differential equations, while Markov processes specify the latter. Thus, the concept of asymptotic stability used to identify the probable outcomes of the replicator dynamics is formulated in the wrong way to pick out the probable outcomes of the Moran process. Indeed, even the Moran process and Moran process with mutation require distinct stability concepts with which to identify their probable outcomes.

As we have seen, the Moran process constitutes a special kind of Markov process—an absorbing chain—where the process ends up, in the long run, in one or another absorbing state. This allows us to assess the behavior of the process in terms of fixation probabilities. With the introduction of mutation, the Moran process becomes a different kind of Markov process—an irreducible chain—where all states are visited infinitely often, in the limit of time. There are no absorbing states in such a process, and so fixation probabilities have no coherent definition. Distinct but intercomparable concepts are needed for the analysis of these dynamics. To be able to effectively contrast the success of equilibrium concepts under the RD, MP and MPM, we need to refer back to of our novel account of evolutionary significance to guide us in determining suitable analogues for the stochastic context.

Recall that we defined in (1.2.4) a set of states as a *probable outcome of evolution* just in case, given a continuous (or exponential) probability distribution over the initial conditions of the state space, a significant set of initial conditions converge to the set in the long run, where they are stable in the face of arbitrarily small perturbations. We will see that these criteria can be translated for the Moran process in terms of protection from invasion and replacement by any mutation strategy (as described by Nowak [2006]), and for the Moran process in terms of stochastic stability (as described by Young [1998]).

3.2.1 Fixations Probabilities, Invasion and Replacement

Nowak provides us with natural conditions for evolutionary significance for the Moran process $\{X_t^N\}$ that are formulated terms of fixation probabilities. These conditions specify whether a given resident strategy is protected by selection against invasions and replacement by any mutant strategy. He captures these conditions in the ESS^N concept.

There are two logical requirements for evolutionary stability in finite populations. Selection has to protect an ESS^N from invasion by any mutant strategy, and selection has to protect an ESS^N from replacement by any mutant strategy. We compare the fitness of each strategy, f_i and g_i, to see which strategy is favored by selection, given that the population is in a state with i A-types. Denote by h_i the difference in fitness between the two types

$$h_i = f_i - g_i, \qquad i = 1,2, \dots, N.$$

Invasion dynamics then can be determined by evaluating the sign of h_1 and h_{N-1}, where h_1 is the difference in fitness between A-types and B-types when there is a single A-type mutant in the population, and h_{N-1} is the difference in fitness when there is a single B-type mutant in the population. We define these as follows

> **3.2.1.1 Definition (Invasion):** If $h_1 < 0$, then selection *opposes* A *invading* B, and if $h_{N-1} > 0$, then *selection opposes* B *invading* A.

In other words, if the fitness of a single mutant is less than the fitness of the resident population, then selection opposes invasion. Conversely, a strategy favored by selection for invading is likely to find a foothold in the population. This does not indicate, however, whether the invasive strategy will ultimately come to take over the population. For this, we turn to the concept of replacement

> **3.2.1.2 Definition (Replacement):** If $\rho_{AB} < \frac{1}{N}$ then *selection opposes* A *replacing* B, and If $\rho_{BA} < \frac{1}{N}$ then *selection opposes* B *replacing* A.

That is, if the fixation probability of the mutant is less than that of a neutral mutant (whose fixation probability is $\frac{1}{N}$), then selection opposes fixation. The first condition $(3.2.1.1)$ is always a simple linear inequality in N, while the second condition $(3.2.1.2)$ is only a linear inequality in N for weak selection $w \to 0$. The conditions for invasion are easy to compute. In contrast, the conditions for replacement employ fixation probabilities, and so often cannot be solved for N in the absence of weak selection. If we do take the limit of weak selection, we can use a Taylor expansion of fixation probabilities to make the calculation of replacement probabilities tractable. Hence, we will proceed with our analysis using the limit of weak selection, and return to relax this assumption.

Following Nowak [2006], we define the two logical conditions for evolutionary stability for the Moran process as follows

> **3.2.1.3 Definition (ESS^N)**
> A is an ESS^N if, for any other strategy $B \in S$,
> (1) selection opposes B invading A, and
> (2) selection opposes B replacing A.

When we consider the limit of weak selection limit $w \to 0$, a Taylor expansion of the equation for the fixation probability of A yields

$$\rho_A \approx \frac{1}{N}\frac{1}{1-\frac{(\alpha N-\beta)w}{6}}.$$

Here, $\alpha = a + 2b - c - 2d$ and $\beta = 2a + b + c - 4d$. If $\rho_A > 1/N$, then selection favors the fixation of A. From the previous equation, it follows that $\rho_A > 1/N$ is equivalent to $\alpha N > B$. This condition can be written as

$$A(N-2) + b(2N-1) > c(N+1) + d(N-4).$$

For a population of just two individuals $N = 2$, the inequality simplifies to $b > c$. And for the large population limit $N \to \infty$, the inequality simplifies to $a + 2b > c + 2d$. This should make intuitive sense: In a population of one A individual and one B individual, the former has a payoff b and latter has a payoff of c. The fixation probability then is wholly determined by the difference between the values of b and c. As the population size increases, the fixation probabilities, and the behavior of Moran process generally, come to be determined by the differences between a and c, and of b and d in a way familiar to us from the replicator dynamics where the latter two differences wholly determine the qualitative behavior of the dynamics.

Qualitative change in the behavior of the Moran Process as a function of population size can be wholly understood in light of the transition between the determination of its behavior by the difference of b and c in a the two-agent population, and of a and c, and of b and d, in the large population limit. This yields another basic insights: Unlike in the replicator dynamics [Cressman, 2003], the qualitative dynamics of the Moran process is not invariant under positive affine transformations of the columns of the payoff matrix. We capture this insight in the following formulations of the ESS^N

3.2.1.4 Definition (ESS^N for small N $[N = 2]$)
A is an ESS^N if (1) $b > c$, and (2) $b > c$.

3.2.1.5 Definition (ESS^N for large N $[N \to \infty]$)
A is and ESS^N if (1) $a > c$, and (2) $x^* > 2/3$.

The last inequality indicates an interesting result, derived by Nowak [2006, Ch.7]: Selection protects any strategy from replacement against strategies whose basin of attraction composes less than $1/3$ of the state space.

We can contrast this with the traditional ESS concept in the RD, which captures the idea that selection opposes the spread of infinitesimally small fractions of

mutants by infinitely large incumbent populations. For our 2x2 symmetric game, the traditional ESS is defined as follows

3.2.1.6 Definition (traditional ESS for 2x2 symmetric games)
A is an ESS if (1) $a > c$, and (2) If $a = c$, then $b > d$.

Thus, for small N, the traditional ESS is neither necessary nor sufficient to imply protection against invasion and replacement. In the limit of large population size and weak selection, however, the ESS provides necessary but not sufficient conditions for stability for the Moran process $\{X_t^N\}$.

Four our purposes, the ESS^N adequately captures the notion of evolutionary significance in the context of the Moran process. First, criterion (1) corresponds neatly to the notion that an evolutionarily significant states is stable in the face of arbitrarily small perturbations. Second, criterion (2) ensures that there is a significant basin of attraction from which the process is likely to converge to the ESS^N state in the limit.

3.2.2 Stochastic Stability

Where invasion and replacement provide concepts to quantify the behavior of the MP $\{X_t^N\}$, stochastic stability can be used to capture the probable outcomes of the Moran process with mutation $\{X_t^{N,\eta}\}$. The concept of stochastic stability was originally introduced to game theory by Foster and Young in [1990]. Recall that, which the introduction of mutation, the Moran process becomes an irreducible Markov process. The perturbed process $\{X_t^{N,\eta}\}$ now visits each state infinitely often in infinite-horizon play. Given this, the best we can do is to find the distribution of time spent over different states in the state space χ^N as given by the unique stationary distribution μ of the process. The states that are stochastically stable are just those that retain mass in the stationary distribution μ. First, we define stochastic stability in the small mutation limit before proceeding to a more general formulation suitable for fixed finite population size N and mutation rate η.

3.2.2.1 Definition (Stochastic Stability for vanishing η)
A state $i \in X^N$ is *stochastically stable* in the small mutation limit if

$$\lim_{\eta \to 0} \mu_i^{N,\eta} > 0$$

When the sequence $\{\mu^{N,\eta}\}_{\eta\in(0,\overline{\eta}]}$ converges to a point mass at state i, state i is *uniquely stochastically stable*.

Questions about stochastic stability can be answered by using particular well-chosen graphs, obviating the need to compute the stationary distribution (see Young [1998, Ch.4] for the relevant techniques).

If values of N and η are held fixed, however, it is not possible to obtain clean selection results. The stationary distribution $\mu^{N,\eta}$ necessarily puts positive mass on every state in X^N. For this context, we define a refined notion of local stochastic stability as follows

> **Definition (Local Stochastic Stability [fixed N and η])** A state i is *locally stochastically stable* for an irreducible Markov process $\{X_t^{N,\eta}\}$, with respect to fixed values of N and η, if and only if, state i is a peak of the stationary distribution in the limit of time $t \to \infty$.

This formulation helps to select the locally probable outcomes of the process amongst the range of potential candidates. We note that the mass in the stationary distribution can only become concentrated around stationary points of the replicator dynamic. Even when the underlying game has multiple strict equilibria, the stationary distribution is often concentrated in the vicinity of just one of them if the population size N and mutation level η are sufficiently small. In these cases, the population state so selected provides a unique prediction of infinite horizon play.

Stochastic stability does amount to a perfect stand-in for our notion of evolutionary significance. But, it is unclear if there are any better tools at our disposal. The good news is that our requirement that an evolutionarily significant state command convergence from a significant basin of attraction is plausibly satisfied. All and only those states that constitute probable locations of the process in the long run have positive mass in the stationary distribution. The bad news, however, is that the requirement that evolutionarily significant states be stable in the face of arbitrarily small perturbations does not look like it can be met. With the introduction of stochastic mutations, all states are vulnerable to small perturbations. Indeed, it is the rare but inevitable series of small mutations strung together that force the motion of the process to be forever an itinerant of the state space. We proceed in our analysis with the concepts of stochastic stability and local stochastic stability as our formal criteria for evolutionary significance for the MPM.

3.3 Results and Analysis

To understand the limitations of equilibrium concepts for our stochastic dynamics, we can leverage our existing insights as to the points of agreement between the replicator dynamics (RD) and Moran process (MP). We examine the behavior of the MP & the Moran process with mutation (MPM) under different values of population size N, and mutation rate η. We maintain an assumption of weak selection $w \to 0$ throughout. Intensity of selection can have qualitative effects of the dynamics, and demands study. We do not explore the effects of selection intensity here, but the assumption of weak selection does not wholly undermine our results. There are a breadth of applications in which weak selection constitutes a tenable assumption. For example, in the application of evolutionary dynamics to biology, most genetic mutations lead to very small changes in the phenotype of organisms, as so have only very small effects on their fitness. The overwhelming majority of genetic mutations do not effect fitness at all and are called 'neutral mutation'. For this reason, many results in population genetics are derived for the limit of weak mutation or even neutrality [Nowak, 2006]. More generally, the interactions considered in individual evolutionary games do not constitute the bulk of where the organisms under consideration derive their fitness. Organisms are typically engaged in many more interactions with other members of their species, with other species, and with nature, that can constitute the core factors determining their reproductive success.

Even in very simple games, the behavior of the MP and MPM with mutation can diverge significantly from those of the RD. Both their points of overlap and divergence are instructive. We begin our analysis with 2x2 symmetric games, and develop out intuitions as to the basic behaviors of these processes before moving on to more complex cases.

First we consider the most analytically tractable cases of Moran process $\{X_t^N\}$ in the large population limit $N \to \infty$, and the Moran process with mutation $\{X_t^{N,\eta}\}$ in the limit of large populations $N \to \infty$ and vanishing mutations $\eta \to 0$. We then consider the case of fixed finite populations N. Finally, we consider the case of fixed finite populations N with fixed nonzero mutation rates $\eta > 0$. In this last case, we discover an interesting and novel result that provides a partial vindication for the predictions of the replicator dynamics. In each case, we assess the success of the equilibrium concepts in capturing the evolutionary significant states indicated by the dynamics and their corresponding stability concepts—the ESS^N and stochastic stability.

3.3.1 Equilibrium Concepts Under the Moran Process

The behavior of the Moran $\{X_t^{N,}\}$ process can be complex, even in the large population limit $N \to \infty$. We follow [Taylor et all, 2004] in leveraging known agreement and divergence between the MP and the RD to deduce essential features of the behaviors of the former by contrasting it with the latter. We begin with similarities. First, any stable state of the MP is known to be a stationary point of the RD [Fudenberg et al, 2006]. Specifically, it will be vertex of the state space corresponding to the pure strategy vector e_i for some $i \in S$. This is because, as we have seen, any stable state of the MP must be a monomorphic state, and all monomorphic states are stationary points of the RD.

If there is only a single monomorphic AS state of the dynamics under RD, it will be stable under the MP [Taylor et al, 2004]. Unlike under the RD in §2, however, this stable state is guaranteed to be a strict Nash equilibrium. This is because, for the MP in the large population limit, we cannot have stable states that are not also Nash. Since stable state must be monomorphic, this rules out traditionally problematic behaviors such as elliptical attractors. Indeed it also rules out other evolutionarily significant behavior under the RD that could not be captured by the equilibrium concepts: AS sets, cycles, and strange attractors. All of these are polymorphic states, and so cannot be stable under the MP.

In summary, globally asymptotically stable monomorphic states of the RD are stable for the MP. But not all AS monomorphic states of the RD are stable under the MP. Elliptical attractors, AS sets, cycles, and chaotic behavior are impossible for the MP since they are polymorphic states. Hence, we can also state the previous insight in terms of equilibrium concepts: All unique strict pure-strategy Nash equilibria are stable under the MP. But not all strict pure-strategy Nash equilibria are stable.

Table 2: 2x2 Symmetric Games under the Replicator Dynamics

Game Type	Payoffs	Phase Portrait	Stationary Points	AS (and ES) states
A Dominates B	$a > c$ $b > d$	A ●⟵○ B	$\Delta^{RP} = \{0, 1\}$	$\Delta^{AS} = \{0\}$
Bi-stable Case	$a > c$ $b < d$	A ●⟵○⟶● B	$\Delta^{RP} = \left\{0, \frac{d-b}{d+a-b-c}, 1\right\}$	$\Delta^{AS} = \{0,1\}$

Polymorphic Case	$a < c$ $b > d$	A ⟵●⟶ B	$\Delta^{RP} = \left\{0, \frac{b-d}{b+c-a-d}, 1\right\}$	$\Delta^{AS} = \left\{\frac{b-d}{b+c-a-d}\right\}$
B Dominates A	$a < c$ $b < d$	A ○⟶● B	$\Delta^{RP} = \{0,1\}$	$\Delta^{AS} = \{1\}$
Neutral Case	$a = c$ $b = d$	A ····· B	$\Delta^{RP} = [0,1]$	$\Delta^{AS} = \emptyset$

We remind ourselves of the five possible qualitative dynamics of symmetric 2x2 games under the RD in TABLE 2 above. In understanding the divergence of the RD and the MP, there are two cases that may be uniquely instructive: the bi-stable case (corresponding to the class of coordination games), and the case where on strategy strictly dominates the other.

TABLE 3: 2x2 Symmetric Games under the Moran Process $\{X_t^N\}$

Game Type	Payoffs	Behavior for $N \to \infty$	Interpretation
A Dominates B	$a > c$ $b > d$	$\rho_{BA} < \frac{1}{N} < \rho_{AB}$ $0 < h_1, h_{N-1}$	Selection opposes B replacing A, and favors A replacing B. Selection opposes B invading A, and favors A invading B.
Bi-stable Case	$a > c$ $b < d$	$h_1 < 0 < h_{N-1}$	Selection opposes invasion by both A and B. But selection may favor either A replacing B, or B replacing A, or neither but not both.
Polymorphic Case	$a < c$ $b > d$	$h_{N-1} < 0 < h_1$	Selection favors invasion by both A and B. But selection may favor either A replacing B, or B replacing A, or both but not neither.
B Dominates A	$a < c$ $b < d$	$\rho_{AB} < \frac{1}{N} < \rho_{BA}$ $h_1, h_{N-1} < 0$	Selection opposes A replacing B, and favors B replacing A. And selection opposes A invading B, and favors B invading A.
Neutral Case	$a = c$ $b = d$	(If $a + c = b + d$) then $\rho_{AB} = \rho_{BA} = \frac{1}{N}$	Selection may favor invasion and replacement by A or B depending on the sign of $(a + c) - (b + d)$, and if $a + c = b + d$, then we have neutral drift

The bi-stable case demonstrates how pure-strategy strict Nash equilibria can be unstable under the MP (TABLE 3). In this case, there are two strict pure-strategy Nash equilibria corresponding to the states all-A and all-B, and one weak Nash

equilibrium at $x^* = \frac{d-b}{d+a-b-c}$. The weak Nash x^* cannot be stable, since it is mixed. We will show that one of the strategies, say A, need not be stable—is not an ESS^N. The first condition for an ESS^N then is that selection oppose B invading A $(a > c)$. This is satisfied by the assumption that this is a coordination game ($a > c$ and $b < d$). Then for B to fail to be an ESS^N, we must be able to construct a game where selection fails to oppose B replacing A. This amounts to B possessing a basin of attraction that composes greater than one third of the state space. Anytime $x^* \leq 2/3$ this is satisfied. And so A is a strict pure-strategy Nash equilibrium, but A can be unstable (not an ESS^N) under the MP, as desired. ■

This demonstrates that if a strict pure-strategy Nash equilibrium is not the unique equilibrium of the game, it need not be stable. Since all of our other equilibrium concepts are weaker than the strict Nash equilibrium (i.e., the set of strict Nash equilibria is a subset of all other equilibria), all of our equilibrium concepts are too weak in this regard. Thus, all of our equilibrium concepts can mistakenly pick out unstable states in the class of games where there is more than one strict pure strategy Nash equilibrium. The qualitative dynamics of the Moran process in the large population limit are given in TABLE 3 above.

TABLE 4 summarizes our assessment of the limitations of the equilibrium concepts under the Moran process in the large population limit. Observe that the Nash equilibrium, and hence all other equilibrium concepts, provide necessary but not sufficient conditions for evolutionary significance. They are all too weak, but none is too strong. It is interesting that, unlike in the RD (TABLE 1), none of the equilibrium concepts is too strong. This can be attributed to the fact that complex evolutionarily significant outcomes, such as elliptical attractors, are ruled out by the fact that long run outcomes of the MP $\{X_t^N\}$ must be monomorphic.

TABLE 4: Equilibrium Concepts for the Moran Process $\{X_t^N\}$ $(N \to \infty)$

	Too Weak Captures some outcomes that ARE NOT evolutionarily significant that are:		**Too Strong** Fails to capture some outcomes that ARE significant that are:
	Not Absorbing States	Not ESS^N	ESS^N
Nash Equilibrium (NE)	☑ *(Mixed equilibria)*	☑ *(Bi-stable case)*	☐

Strict Nash Equilibrium (Strict NE)	☑ *(Mixed equilibria)*	☑ *(Bi-stable case)*	☐
Evolutionarily Stable Strategy (ESS)	☑ *(Mixed equilibria)*	☑ *(Bi-stable case)*	☐
Neutrally Stable State (NSS)	☑ *(Mixed equilibria)*	☑ *(Bi-stable case)*	☐
Evolutionarily Stable Set (ESSet)	☑ *(Mixed equilibria)*	☑ *(Bi-stable case)*	☐

3.3.2 Equilibrium Concepts Under the Moran Process With Finite Population

To understand the complexities introduced by finite population dynamics for the MP, the dominated strategy case is instructive. When we relax the assumption of the large population limit, the MP undergoes qualitative shifts in its behavior that are a function of population size. Now, even *unique* pure-strategy strict Nash equilibria need not be stable. And the equilibrium concepts can provide neither necessary nor sufficient conditions for evolutionary significance.

In the large population limit, the dynamics for the game where strategy A dominates strategy B behave as expected. Namely, selection opposes B and favors A with respect to both invasion and replacement. However, as we have seen, for $N = 2$, the strictly dominated strategy can fair well as long as $c > b$. That is, there is spite. In fact, it is easily observable that the direction of selection with respect to invasion and replacement is gradually inverted. Consider the following example proffered by Taylor et al [2004].

$$\Gamma = \begin{pmatrix} 3.1 & 1.02 \\ 3 & 1 \end{pmatrix}$$

For this game Γ, where strategy A strictly dominates strategy B, the Moran process $\{X_t^N\}$ expresses qualitatively distinct dynamics as a function of population size. This is illustrated in TABLE 5 bellow. Observe the stepwise inversion of the replacement and invasion dynamics.

TABLE 5: Game Γ under the MP with different population sizes *N*

$N \leq 21$	$21 < N \leq 30$	$30 < N \leq 50$	$50 < N \leq 101$	$101 < N$
$\rho_{AB} < \frac{1}{N} < \rho_{BA}$ $h_1, h_{N-1} < 0$	$\rho_{AB} < \frac{1}{N} < \rho_{BA}$ $h_1 < 0 < h_{N-1}$	$\rho_{AB}, \rho_{BA} < \frac{1}{N}$ $h_1 < 0 < h_{N-1}$	$\rho_{BA} < \frac{1}{N} < \rho_{AB}$ $h_1 < 0 < h_{N-1}$	$\rho_{BA} < \frac{1}{N} < \rho_{AB}$ $0 < h_1, h_{N-1}$

More generally, we can be certain that if $c > b$, then there is a population size $N_o \geq 2$ such that for all $N < N_o$ we have $\rho_{AB} < \frac{1}{N} < \rho_{BA}$ [Taylor et al, 2004]. The moral is that, for finite population sizes, the probable outcomes of evolution can undergo complete inversions as a function of population size, and so equilibrium concepts insensitive to this parameter have little hope of providing accurate predictions.

3.3.3 Equilibrium Concepts Under the Moran Process With Mutations

If the behavior of the Moran process $\{X_t^N\}$ is complex, the behavior of the Moran process with mutation $\{X_t^{N,\eta}\}$ is even more so. In the limit of large population $N \to \infty$ and vanishing mutation $\eta \to 0$, the set of evolutionarily significant states of the MPM are a subset of the absorbing states of the MP. Just as we leveraged the RD to better understand the MP, we will use the behavior of the MP to gain insight into the behavior of the MPM. This is particularly appropriate when we take the small mutation limit. Intuitively, this is because the behavior of the two dynamics becomes more similar as mutation vanishes.

We leverage known agreement and divergence between the MP and the MPM to deduce essential features of the behaviors of the former by comparing it to the latter. We begin with similarities. First, in the small mutation limit, any stochastically stable state of the MPM must be an absorbing state of the MP. But typically only a single state of the MPM may be stochastically stable. If there are multiple absorbing states, then the absorbing state with the greatest fixation probability under the MP will be the stochastically stable one [Taylor et al, 2004]. Only in the knife-edge cases of multiple absorbing states with equal fixation probabilities can we get multiple stochastically stable states.

Importantly, stochastically stable states of the MPM need not be stable (ESS^N) states of the MP. There will always be at least one stochastically stable state of a game. This is because the stochastically stable state is determined by having

nonzero mass in the stationary distribution μ. And, as a probability distribution, the stationary distribution μ must assign nonzero mass to at least one state.

The divergence between stochastically stable states of the MPM and stable states of the MP occurs in games where there are no stable states in the MP. We will shortly consider the example of anti-coordination games. In such a game, there can be a single stochastically stable state that is not an ESS^N.

In summary, the set of stochastically stable states of the MPM are a subset of the absorbing states of the MP. When there is a single stable state of the MP, this will be stochastically stable under the MPM. But stochastically stable states of the MPM need not be stable states of the MP. This implies that globally asymptotically stable monomorphic states of the RD are also stable for the MPM. But not all AS monomorphic states are stable for the MP. Polymorphic states still cannot be stable. We can formulate the previous insight in terms of equilibrium concepts: All unique strict pure-strategy Nash equilibria are stable under the MP. But not all strict pure-strategy Nash equilibria are stable. Moreover, not all stochastically states are even Nash equilibria. The qualitative dynamics of the MPM in the large population limit are given in TABLE 6 bellow.

TABLE 6: 2x2 Symmetric Games under the Moran Process with Mutation $\{X_t^{N,\eta}\}$

Game Type	Payoffs	Behavior for $N \to \infty, \eta \to 0$	Interpretation
A Dominates B	$a > c$ $b > d$	$\mu_A \to 1$	The stationary distribution μ will be concentrated in a point-mass at all-A.
Bi-stable Case	$a > c$ $b < d$	Either $\mu_A \to 1$ or $\mu_B \to 1$	The stationary distribution μ will be concentrated in a point-mass at either all-A or all-B.
Polymorphic Case	$a < c$ $b > d$	Either $\mu_A \to 1$ or $\mu_B \to 1$	The stationary distribution μ will be concentrated in a point-mass around either all-A or all-B. (Note: this is the same, qualitatively, as the bi-stable case)
B Dominates A	$a < c$ $b < d$	$\mu_B \to 1$	The stationary distribution μ will be concentrated in a point-mass at all-B.
Neutral Case	$a = c$ $b = d$	If $a + c = b + d$ then $\mu_A = \mu_B = \frac{1}{2}$	The process may spend its time overwhelming at A or B depending on the sign of $(a + c) - (b + d)$, or equally between them if $a + c = b + d$.

For the MPM, it is the polymorphic case (corresponding to the class of anti-coordination games) that is particularly instructive. The polymorphic case

demonstrates how non-Nash states can be stochastically stable under the MPM $\{X_t^{N,\eta}\}$. In this case, there is only one symmetric Nash equilibrium—the mixed equilibrium at $x^* = \frac{b-d}{b+c-a-d}$. But as we know, as a non-monomorphic state, x^* cannot be stochastically stable. But, one of the monomorphic stationary points, corresponding to the all-A and all-B population states, can indeed be stochastically stable. We will show that one of the strategies, say A, can be stochastically stable. The condition for stochastic stability for such a games reduces to determining which strategy has a greater fixation probability under the MP. Let γ denote the ratio of the fixation probabilities $\frac{\rho_{AB}}{\rho_{BA}} = \prod_{k=1}^{N-1} \frac{f_k}{g_k}$. So, if $\gamma > 1$, then A is stochastically stable. This is equibalent to satisfying condition (2) of the ESS^N: that $x^* > 2/3$. This can clearly be satisfied. Hence, A is not a Nash equilibrium, but A is stochastically stable under the MPM, as desired. ■

This demonstrates that stable states of the MPM need not be Nash equilibria. Since all of our equilibrium concepts are stronger than the Nash equilibrium (so that the set of Nash equilibria is a superset of the set of all of our other equilibrium concepts), we can conclude that the equilibrium concepts are all too strong in this respect. All our equilibrium concepts can fail to capture stable states in the class of games where the unique Nash equilibrium is a polymorphic state.

TABLE 7 summarizes our assessment of the limitations of the equilibrium concepts under the Moran process with mutation, in the large population and small mutation limits. Observe that the equilibrium concepts provide neither necessary nor sufficient conditions for evolutionary significance here. They are all simultaneously too weak and too strong.

TABLE 7: Equilibrium Concepts for the Moran Process with Mutation $\{X_t^{N,\eta}\}$ ($N \to \infty, \eta \to 0$)

	Too Weak Captures some outcomes that ARE NOT evolutionarily significant that are:	**Too Strong** Fails to capture some outcomes that ARE evolutionarily significant that are:
	Not Stochastically Stable	Stochastically Stable
Nash Equilibrium (NE)	☑ *(Bi-stable case)*	☑ *(Polymorphic case)*
Strict Nash Equilibrium (Strict NE)	☑ *(Bi-stable case)*	☑ *(Polymorphic case)*

Evolutionarily Stable Strategy (ESS)	☑ *(Bi-stable case)*	☑ *(Polymorphic case)*
Neutrally Stable State (NSS)	☑ *(Bi-stable case)*	☑ *(Polymorphic case)*
Evolutionarily Stable Set (ESSet)	☑ *(Bi-stable case)*	☑ *(Polymorphic case)*

3.3.4 Equilibrium Concepts Under the Moran Process With Mutation and Finite Population

When we relax the assumption of the large population limit, the MPM also undergoes qualitative shifts in its behavior that are contingent on population size. As a result, we can get inversions of the dynamics analogous to those of the MP. A strategy that is stochastically stable for small populations can become unstable as the population size increases, and vice versa.

A new factor in understanding stochastic stability in these contexts is that the stationary distribution no longer (typically) picks out a single state to which to assign nonzero probability. The fact that only a single state is selected by the stationary distribution is, in a sense, an artifact of the assumption of vanishing mutation rates. Indeed, since the process is irreducible, the stationary distribution for fixed nonzero N and η will assign mass to every state in the finite state space. Under these conditions, in order to pick out the most probable outcomes of evolution, and to summarize the properties of such a distribution, we must use our alternative formulation of stochastically stability: local stochastic stability—where the peaks of the stationary distribution are selected.

We can revisit the game Γ from §3.2.2, where strategy A strictly dominates strategy B. Under the MPM with vanishing mutation, we see qualitative shifts in the dynamics as a function of the population size.

Table 8: Game Γ under the MPM with different population sizes N

$N = 2$	$N < 42$	$N = 42$	$42 < N$	$N \to \infty$
$\mu_B \approx 1$	$\mu_A < \mu_B$	$\mu_A \approx \mu_B \approx 1/2$	$\mu_A > \mu_B$	$\mu_A \to 1$

3.3.5 Equilibrium Concepts Under the Moran Process With Fixed Population Size and Mutation Rate

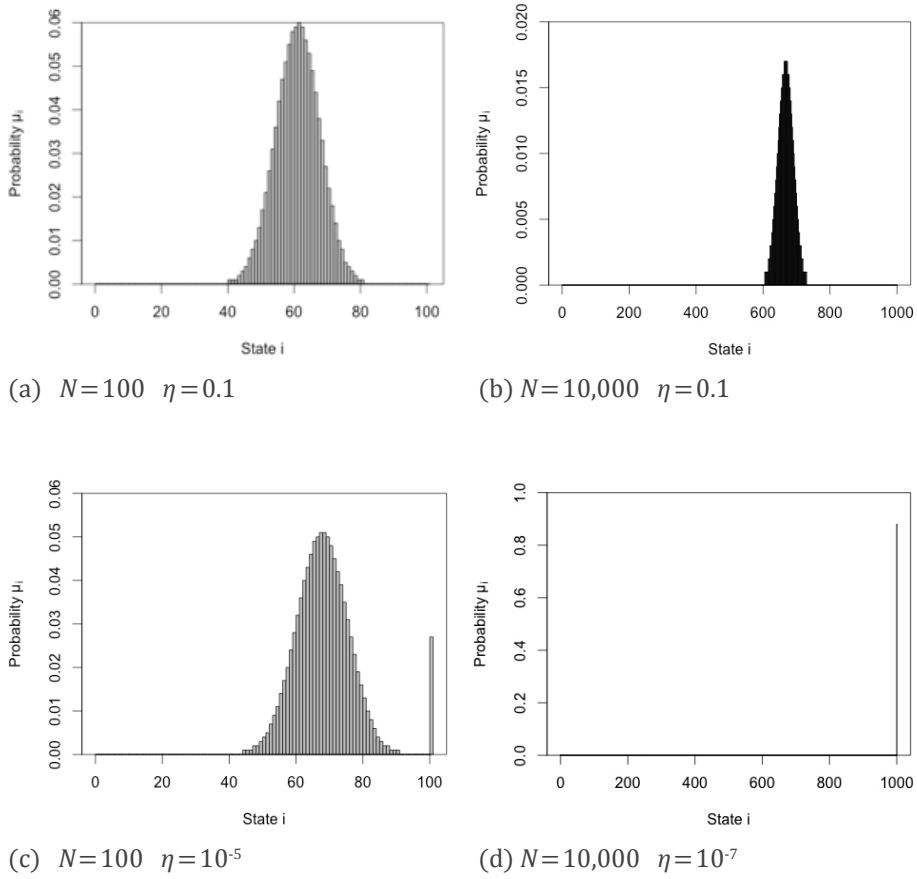
When we relax both the assumptions of infinite populations and vanishing mutation, the MPM exhibits a confounding diversity of qualitative behaviors. When population is sufficiently large and mutation sufficiently rare, we retrieve the limit behavior previously analyzed. But for the breadth of intermediate values, which can include populations that are quite large, and mutation rates that are quite small, the general behavior of the MPM is not captured by its limit behavior. Perhaps the most interesting result is that, for a large range of values of population size N and mutation rate η, the behavior of the MPM accords strikingly with that of the RD. We will discover that strong mutation $\eta N > 1$ provides a sufficient condition for the qualitative agreement in the behavior of the MPM and RD for the class of 2x2 symmetric games, and possibly more broadly. When this agreement is in place, the fate of the equilibrium concepts under the MPM corresponds to that predicted under RD.

This agreement between the RD and MPM should not be surprising: As we have discussed, the RD provides the mean field of the MP. But what is formally known about this relationship is that, in the large population limit $N \to \infty$, the MP will, with high probability, remain close to the path of the RD for some large but finite time T [Weibull, 2003]. The domain of correspondence we have discovered is distinct from this. This new domain involves fixed finite population sizes $N < \infty$ and nonzero mutation rates $\eta > 0$ that satisfy the inequality $\eta N > 1$, in the limit of time $t \to \infty$.

The complexity of the MPM for fixed values of N and η typically makes the use of analytic methods impractical. Calculating the fixation probabilities or stationary distribution for fixed N and η is generally a difficult task [Young, 1993; 1998]. Hence, we must turn to computational methods to obtain informative approximations, and to simulations to probe the behavior of the dynamics. With the help of these, we attempt to ascertain the conditions for agreement between the MPM and RD.

To understand the qualitative shifts of the MPM as a function of population size N and mutation rate η, consider the behavior of the dynamics, as described by the stationary distribution, under the following anti-coordination game

$$\Gamma = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$$



(a) $N=100 \quad \eta=0.1$ (b) $N=10{,}000 \quad \eta=0.1$

(c) $N=100 \quad \eta=10^{-5}$ (d) $N=10{,}000 \quad \eta=10^{-7}$

FIGURE 3.1: *Stationary distribution weight* μ *in the anticoordination game* Γ *under the Moran process with mutation for different population sizes* N *and mutation rates* η.

In FIGURE 3.1.a, we see that a small population size of $N = 100$ and very high mutation rate of $\eta = 0.1$ yields behavior that does not agree with the prediction of the vanishing mutation limit. Indeed, the distribution peaks at the asymptotically stable state $x = 2/3$ of the replicator dynamics for the same game Γ. As we increase the population to $N = 10{,}000$, the distribution begins to collapses around x (FIGURE 3.1.b). For larger population sizes generally, the mutation rates at which the process is faithful to the behavior of the replicator dynamics can be even smaller. But, when we hold the population size fixed at $N = 100$ and the lower mutation rate to $\eta = 10^{-5}$, the process begins to put weight on the absorbing state of the MP favored by selection, where the entire population is A-type (FIGURE 3.1.c). Finally, in FIGURE 3.1.d, we see that as we move toward the small mutation limit with $\eta = 10^{-7}$ the stationary distribution becomes concentrated on the boundary states regardless of the behavior of the replicator dynamics.

In the context of cultural evolution, the frequency of mutation may not be small, and so limit methodology with respect to small mutations may yield qualitatively inaccurate predictions. Even in biological contexts, for sufficiently large populations, the Moran process can remain in agreement with the replicator dynamics when mutations are rare. We are considered to be working with strong mutation when $\eta N > 1$ [Traulsen & Hauert, 2009]. Intuitively, this is where at least one new mutation enters the population per generation.

Our computational results indicate that for such rates of mutation the behavior of the Moran process consistently accords with that described by the replicator dynamics. Consider bacterial populations, where the per-site mutation rate is of the order 10^{-10} per nucleotide [Barrick et al, 2009], translating to a per-genome mutation rate (for E. coli bacteria) of about 4×10^{-4}. That is, only 4 out of $10{,}000$ bacteria produced carry at least one mutation. Typical bacterial populations, however, are of the order of $10^6 - 10^8$. This yields mutation strength of $\eta N \approx 20{,}000$. There will be an average of roughly twenty thousand mutations per generation. This is deep into the territory of strong mutation. Hence, bacterial populations express mutation rates at odds with the assumption of the weak mutation limit.

What we have then is the interesting result that strong mutation $(\eta N > 1)$ seems to provide a sufficient condition for the agreement of the Moran process with the replicator dynamics under 2x2 symmetric normal form games.

> **Claim (MPM-RD harmony for strong mutation)** For any 2x2 symmetric games Γ, in the presence of strong mutation $\eta N > 1$, a state i is a peak of the stationary distribution μ of the Moran process with mutation just in case it is asymptotically stable under the replicator dynamics for the same game.

Given that we know that the ESS captures all and only the evolutionarily significant states of 2x2 symmetric games under the replicator dynamics, it follows that the ESS captures all and only the significant states of the MPM in the presence of strong mutation. What we find here is an unexpected window of success for the ESS concept.

Moreover, computational results of the dynamics of the RPS game ($Figure\ 3$) and Zeeman's elliptical attractor ($Figure\ 11$) indicate that this relation may hold for 3x3 symmetric games. We conjecture that the agreement may obtain for all interaction structures. Exploration of the classes of games under which strong mutation provides a sufficient condition for agreement between the MPM and RD constitutes an interesting subject for future study. However, computational

results also indicate that processes with less than strong mutation rates can also express agreement between the dynamics. (This can be detected by the persistence of stable polymorphic populations.) This implies that strong mutation, while sufficient, is not necessary. It would be valuable, if possible, to find necessary and sufficient conditions for this coincidence.

TABLE 9 summarizes our assessment of the limitations of the equilibrium concepts under the Moran process for strong selection $\eta N > 1$. Observe that, due to the agreement of the MPM and the RD in this case, the equilibrium concepts perform just as they do under the RD.

TABLE 9: Equilibrium Concepts *for* Moran Process with Mutation (for strong selection $\eta N > 1$)

	Too Weak Captures some outcomes that ARE NOT evolutionarily significant that are:		**Too Strong** Fails to capture some outcomes that ARE evolutionarily significant that are:
	Not Locally Stochastically Stable	Small Basin of Attraction	Locally Stochastically Stable
Nash Equilibrium (NE)	☑ *(Non AS states)*	☑ *(Bi-stable case)*	☑ *(Elliptical attractors)*
Strict Nash Equilibrium (Strict NE)	☐	☑ *(Bi-stable case)*	☑ *(Elliptical attractors)*
Evolutionarily Stable Strategy (ESS)	☐	☑ *(Bi-stable case)*	☑ *(Elliptical attractors)*
Neutrally Stable State (NSS)	☑ *(Non AS states)*	☑ *(Bi-stable case)*	☑ *(Elliptical attractors)*
Evolutionarily Stable Set (ESSet)	☐	☑ *(Bi-stable case)*	☑ *(Elliptical attractors)*

3.4 Summary and Discussion

In summary, we have seen that in the simplest and most analytically tractable case—for the Moran process (MP) in the limit of large population size and weak selection—the equilibrium concepts provide sufficient but not necessary conditions for stability. When there is a single monomorphic stable state of the Moran process that is also asymptotically stable under the replicator dynamics, this will be a strict Nash equilibrium. Thus, this state will be correctly predicted by each of the other equilibrium concepts, as the set of strict Nash equilibria composes a (strict) subset of the states picked by each of the other equilibrium concepts. However, not all monomorphic AS states under the RD are stable under the MP. When the are multiple monomorphic AS states of the game under the RD, the equilibrium concepts will not be able to discern which states are stable and which are not, and will make false positive predictions (see the bi-stable case in TABLE 3). These false positives will be strict Nash equilibria. Since the set of strict Nash composes a subset of the or all other equilibrium concepts, this shortcoming is common to all equilibrium concepts. The equilibrium concepts do not make any false negative predictions for the MP, however. This is attributable to the fact that behaviors that were non-Nash AS states under the replicator dynamics—elliptical attractors, AS sets, cycles, and strange attractors—cannot persist in the Moran process. Stable states of the Moran process must be monomorphic. This implies that non-singleton AS sets, cycles, and chaos can constitute only transient behavior of the process.

The Moran process with mutation (MPM) yields similar asymptotic behavior to the MP in the limit of large population and vanishing mutation. The set of states that is stochastically stable in the MPM is a strict subset of the absorbing states of the MP. If there is a single stable state of the MP, this will be the sole stochastically stable of the MPM. If there are multiple stable states of the MP, the state with the highest fixation probability will be stochastically stable for the MPM. There can be more than one stochastically stable state only in cases of perfect ties in fixation probabilities between multiple states. This is because infinite-horizon play amplifies the differences between the stable states to the point where the process ends up spending nearly all of its time at the single state with the strongest basin of attraction. With respect to the equilibrium concepts, this means that they fair strictly worse in terms of making false positive predictions as to the behavior of evolution as given by the MPM. Moreover, there may be stochastically stable states that are even Nash. Intuitively, this is because in the limit of vanishing mutation, the process will still end up spending the majority of its time at the monomorphic states that would be absorbing in the unperturbed process, and these states may not be symmetric Nash equilibria (see the polymorphic case in TABLE 6). Just as in the MP, elliptical attractors, non-singleton sets of AS states,

cycles, and strange attractors behavior cannot be outcomes of the MPM in the limit of vanishing mutations. The stationary distribution of the process assigns zero mass to polymorphic states, and so they cannot be stochastically stable.

A major qualitative shift occurs when we model fixed population sizes N and mutation rates η. For strong mutation $\eta N > 1$, our computations of the stationary distribution of the MPM and simulation results indicate that the process behaves as described by the replicator dynamics for the case of 2x2 symmetric games, and for 3x3 games with elliptical attractors ($Figure\ 11$) and cycles ($Figure\ 3$). This suggests a satisfying harmony between the predictions of the Moran process with mutation with those of the replicator dynamics in the presence of strong selection. But, this does not guarantee that behavior of the two dynamics will not diverge in other game structures. The question as to the limits of the agreement of the dynamics for the case of strong selection provides an interesting topic for future study.

These results raise concerns for limit methodology. In particular, it seems that, in the stochastic context, the assumption of vanishing mutations makes evolutionary behavior—such as stable polymorphic populations—that are readily observable in nature impossible [Sinervo, 1996]. The idealization of infinite populations does not seem to be problematic in the same way. This is because, for sufficiently large populations, the qualitative behavior of the process is identical to that of the infinite population model. This is not so for taking the limit as the mutation rate goes to zero. When mutation vanishes, so does a wealth of phenomena—from interior attractors to cycles—that characterize the behavior of the process at even very small mutation rates. In contrast, the replicator dynamics, which also assume the absence of mutation, do not express the same shortcoming. This may indicate a methodological moral: The formulation of selection in stochastic terms uniquely demands the introduction of the complimentary stochastic force of mutation.

4 Conclusion

Our analysis of the results circumscribing the limitations of equilibrium concepts leaves us in an interesting place. With our novel formulation of evolutionary significance, we have made explicit the interrelationships between evolutionary significance, stability concepts, and equilibrium concepts. In assessing equilibrium concepts, we have seen how stability concepts stand in for evolutionary significance. The determination of the agreement of an equilibrium concept with a stability concept is the proxy for the determination of the agreement of the equilibrium concepts with evolutionary significance. In addition, we have distinguished two distinct arguments against the exclusive use of equilibrium concepts. The first is composed of the demonstration that, even under ideally favorable assumptions, each equilibrium concepts is simultaneously too strong and too weak. The second is an in-principle argument that observes that there are evolutionarily significant outcomes that cannot be effectively formulated in terms of points or sets, and so cannot even be effectively expressed by equilibrium concepts.

In extending our analysis to the stochastic context, we find that our equilibrium concepts fair progressively worse as we relax idealizing assumptions as to the size of populations and the presence of mutations. Unexpectedly, we have found a range of cases, characterized by the presence of strong mutation, under which the equilibrium concepts perform under the Moran process just as they do under the replicator dynamics. Yet, no matter which dynamics we choose, we are faced with the reality that our equilibrium concepts fall short of their aim of predicting the probable outcomes of evolutionary processes.

We have not proven that an equilibrium concept at the level of the payoff matrix must fail in capturing all and only evolutionarily significant outcomes, but we have provided strong arguments indicating that we should not expect a member of the family of Nash refinements can do the job. Our moral is that equilibrium concepts on their own are typically going to be unreliable tools for the analysis of dynamic processes, and that a more complicated, but also more interesting picture emerges from explicit investigation of the underlying dynamics.

Appendix: Algorithms in R

```
# Computing The Moran Process (MP)
# And Moran Process with Mutation (MPM)
# For 2x2 Symmetric Games

# ***Note: Load the 'expm' package for Matrix exponentiation.***

# Let G be a 2x2 symmetric game
# with two players: {1, 2}
# and two pure strategies: {A, B}
# and payoff matrix M = matrix(c(a,b,c,d), nrow=2, ncol=2,
byrow=TRUE).
# Then the payoff to player 1 for each combination of actions by
both players is given by
# U(A,A)=a, U(A,B)=b, U(B,A)=c, U(B,B)=d.
# By symmetry, these also give the payoffs to player 2.

# We assign payoff values for different game types as follows:
assignVec <- Vectorize("assign", c("x","value"))
# Neutral Case (0) : a=c & b=d
#assignVec(c('a','b', 'c', 'd'), c(0, 0, 0, 0), envir
= .GlobalEnv)
# A dominates B (PD) : a>c & b>d
# assignVec(c('a','b', 'c', 'd'), c(2, 0, 3, 1), envir
= .GlobalEnv)
# Bi-stable Case (SH) : a>c & b<d
# assignVec(c('a','b', 'c', 'd'), c(3, 0, 2, 2), envir
= .GlobalEnv)
# Polymorphic Case (HD) : a<c & b>d
# assignVec(c('a','b', 'c', 'd'), c(0, 3, 1, 2), envir
= .GlobalEnv)

# Example: An Anti-coordination Game
 assignVec(c('a','b', 'c', 'd'), c(1, 3, 2, 1), envir
= .GlobalEnv)

# Population Size (we are interested in 'Large Populations' where
N -> ∞)
N <- 1000

# i = number of A-types
# N-i = number of B-types

# Pr(A-type interacts with A-type) = (i-1) / (N-1)
# Pr(A-type interacts with B-type) = (N-i) / (N-1)
# Pr(B-type interacts with A-type) = (i) / (N-1)
# Pr(B-type interacts with B-type) = (N-i-1) / (N-1)

# Expected Payoff to Each Strategy
FA <- function(i) (a*(i-1) + b*(N-i)) / (N-1)
FB <- function(i) (c*(i) + d*(N-i-1)) / (N-1)
```

```
# Intensity of Selection (we are interested in 'Weak Selection'
where w -> 0)
w <- .001

# Fitness (Reproductive/Imitative Success) of Each Strategy
fA <- function(i) (1 - w + w * FA(i))
fB <- function(i) (1 - w + w * FB(i))

# Transition Probabilities for the Moran Process (MP)
# Population Fitness
P <- function(i) (i * fA(i) + (N-i) * fB(i))
# Pr(j -> j+1)
Pforward <- function(i) (i * fA(i) / P(i)) * ((N-i) / N)
# Pr(j -> j-1)
Pback <- function(i) ((N-i) * fB(i) / P(i)) * (i / N)
# Pr(j -> j)
Pstay <- function(i) 1 - (Pforward(i) + Pback(i))

# Mutation Rate (we are interested in 'Vanishing Mutation' where
η->0)
η <- 0.001

# Transition Probabilities for the Moran Process with Mutation
(MPM)
# Pr(j -> j+1)
PforwardM <- function(i) (1-η)*((i * fA(i) / P(i)) * ((N-i) / N))
+ η *(((N-i) * fB(i) / P(i)) * ((N-i) / N))
# Pr(j -> j-1)
PbackM <- function(i) (1-η)*(((N-i) * fB(i) / P(i)) * (i / N)) + η
*((i * fA(i) / P(i)) * (i / N))
# Pr(j -> j)
PstayM <- function(i) 1 - (PforwardM(i) + PbackM(i))

# The MP Transition Matrix := [MP]
MP <- matrix(nrow = N+1, ncol = N+1, byrow = TRUE)

MP <- outer(0:N, 0:N, FUN=function(r,c)
  ifelse(c == r-1, Pback(r),
         ifelse(c == r, Pstay(r),
                ifelse(c == r+1, Pforward(r),
                       0))))

# We re-label the rows and columns to correspond to the number of
A-types in the N-size population.
rownames(MP) = c(0:N)
colnames(MP) = c(0:N)

# The MPM Transition Matrix := [MPM]
MPM <- matrix(nrow = N+1, ncol = N+1, byrow = TRUE)

MPM <- outer(0:N, 0:N, FUN=function(r,c)
  ifelse(c == r-1, PbackM(r),
         ifelse(c == r, PstayM(r),
                ifelse(c == r+1, PforwardM(r),
                       0))))
```

```
# We re-label the row and columns to correspond to the number of
A-types in the N-size population.
rownames(MPM) = c(0:N)
colnames(MPM) = c(0:N)

# We approximate MP^∞ in order to estimate the Reaching
Probabilities from each of the initial conditions to the absorbing
states.
# Denote MP^∞ := MPlim
MPlim <- MP %^% 100000
# We label the matrix, and format P∞ to round numbers to 3 digits.
print("The Moran Process Under the 2x2 Symmetric Game M")
round(MPlim, digits = 3)

# We compute the invasion probabilities for each strategy.
# We compare the fitness of each strategy, fA(i)  and fB(i), to
see which strategy is favored by selection given that the
population is in a state with i A-types. We define this as
follows:
h <- function(i) (fA(i)-fB(i))
# Invasion dynamics then can be determined by evaluating the sign
of h(1) and h(N-1), where h(1) is the difference in fitness
between A-types and B-types where there is a single A-type mutant
in the population, and h(N-1) is the difference in fitness when
there is a single B-type mutant in the population.
# If the fitness of a single mutant is less than the fitness of
the resident population, then selection opposes invasion.
print("Invasion dynamics for A:")
print(h(1))
      if (h(1)>0) {print("Selection favors A invading B")} else
      if (h(1)<0) {print("Selection opposes A invading B")} else
      {print("Invasion is Selection regarding invasion by A")}
print("Invasion dynamics for B:")
print(h(N-1))
      if (h(N-1)>0) {print("Selection opposes B invading A")}
      else if (h(N-1)<0) {print("Selection favors B invading A")}
      else {print("Selection is neutral regarding invasion by
      B")}

# Next, we compute the fixation probabilities, ρAB and ρBA,
corresponding to the probability that the process transitions from
a single A-type mutant to an all-A population, and that the
process transitions from a single B-type mutant to an all-B
population, respectively.
# (To do this, we create a vector of the ratios of the fixation
probabilities fB(i), fA(i), and proceed to compute the cumulative
sum of the cumulative products of transitions over all population
states.)
X <- c(1:N)
HBA <- function(i) (fB(i)/fA(i))
HBAvector <- sapply(X, HBA)
HBAproductsVector <- cumprod(HBAvector)
HBAsumsOfProductsVector <- cumsum(HBAproductsVector)
      ρAB <- 1/(1 + HBAsumsOfProductsVector[N-1])

Y <- c(1:N)
```

```
HAB <- function(i) (fA(i)/fB(i))
HABvector <-sapply(Y, HAB)
HABproductsVector <- cumprod(HABvector)
HABsumsOfProductsVector <- cumsum(HABproductsVector)
       ρBA <- 1/(1 + HABsumsOfProductsVector[N-1])

# We compare the fixation probabilities of each strategy to that
of a neutral mutant (whose fixation probability is 1/N). If the
fixation probability of the mutant is less than that of a neutral
mutant, then we say selection opposes fixation. If the fixation
probability is greater than that of a neutral mutant, then we say
selection favors fixation.
print("Replacement dynamics for A:")
print(ρAB)
       if (ρAB > (1/N)) {print("Selection favors A replacing B")}
       else if (ρAB < (1/N)) {print("Selection opposes A replacing
       B")} else {print("Selection is neutral regarding
       replacement by A")}
print("Replacement dynamics for B:")
print(ρBA)
       if (ρBA > (1/N)) {print("Selection favors B replacing A")}
       else if (ρBA < (1/N)) {print("Selection opposes B replacing
       A")} else {print("Selection is neutral regarding
       replacement by B")}

# We approximate MPM^∞ in order to estimate the Stationary
Distribution μ, which will be the vector of every row of MPM^∞.

# Note: We know that the stationary distribution μ exists because
the addition of mutation makes the Markov process Ergodic. That
is: Where the Moran Process is already finite, and aperiodic, it
is now irreducible (has only one recursive class) as every state
communicates with every other, and so every state will be visited
an infinite number of times in the limit. It follows from being
ergodic that the limit distribution μ is independent of any
initial distribution π(0).

# Denote MPM^∞ := MPMlim
MPMlim <- MPM %^% 100000
# We label the matrix, and format P∞ to round numbers off at the
3rd digit.
print("The Moran Process with Mutation Under the 2x2 Symmetric
Game M")
round(MPMlim, digits = 3)

# Finally, we print the stationary distribution μ for the MPM:
μ <- round(MPMlim[round(N/2),], digits=3)
print("Stationary Distribution μ of the MPM")
print(μ)

# We can create a plot of the stationary distribution μ.
barplot(μ, main="Stationary Distribution μ of the MPM", col =
"gray", ylab=expression('Probability μ'[i]), xlab="State i",
ylim=c(0,1), space=0, xaxt='n')
box()
```

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